

SCI GAME: Let Agents Play the Scientific Discovery Games in the Real and Counterfactual World

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Abstract

Large language model agents are increasingly used for scientific discovery. However, existing benchmarks mix law inference with programming, score data fitting instead of prediction on unseen inputs, and do not separate empirical evidence from pretrained priors. We introduce SCI GAME, a controlled benchmark that converts scientific literature into structured scientific cards. Each card yields a pair of Original and dimensionally consistent Counterfactual environments with the same task interface. SCI GAME contains a Historical collection of canonical laws and a live Temporal collection curated from papers published after model knowledge cutoffs. It evaluates agents in candidate-guided Identification and open-ended Discovery settings. Each setting runs under Prior-only, Fixed, and Adaptive protocols. Experiments on five frontier models yield three findings. First, familiar priors hurt performance once the governing relation is altered. Second, adaptive experimentation helps the strongest models but harms weaker ones under limited budgets. Third, good explanations can hide quantitative failures. Recovering the exact symbolic form remains the main bottleneck, even when agents interpret evidence correctly. Current agents can investigate systems actively, but inducing exact laws in unfamiliar settings remains an open challenge.

1 Introduction

Scientific discovery is a sequential process in which researchers formulate hypotheses, design experiments, interpret observations, and draw conclusions under specified conditions. LLMs increasingly support this process. They analyze scientific descriptions (M. Bran et al., 2024; Laurent et al., 2024), construct executable expressions (Shojaee et al., 2025; Lu et al., 2024), interact with simulated environments (Wang et al., 2022), and propose experiments to test hypotheses (Boiko et al., 2023;

Huang et al., 2025; Gottweis et al., 2026).

Scientific law discovery requires more than fitting data (Schmidt and Lipson, 2009; Brunton et al., 2016; Udrescu and Tegmark, 2020). An agent needs to infer how inputs determine observables and generalize to unqueried inputs (Kronberger et al., 2022). It needs to separate evidence from prior knowledge (Kronberger et al., 2022; Ríos-García et al., 2026), revise hypotheses that contradict observations (Wilie et al., 2024; Jhaveri et al., 2026), and explain the relation and its applicability conditions. A benchmark should therefore evaluate the whole investigation (Ríos-García et al., 2026).

Existing benchmarks cover parts of this process. Some evaluate experiment selection and hypothesis revision interactively (Jansen et al., 2024; Bertolazzi et al., 2026; Chen et al., 2025a). Some recover equations from data (Shojaee et al., 2025) or link modeling with active experimental design (Gandhi et al., 2025; Kabra et al., 2026). Others test reasoning under altered physical laws (Koblischke et al., 2025; Chen et al., 2025b; Zheng et al., 2026; Wiemann et al., 2026) or build tasks from publications (Majumder et al., 2025; Chen et al., 2025c; Starace et al., 2025; Huang et al., 2026a).

Four gaps remain. First, benchmarks often change the task, observation protocol, and metric together. Few compare prior-only reasoning, fixed observations, and adaptive experimentation on one interface, or pair original and counterfactual relations (Chen et al., 2025b; Zheng et al., 2026; Wiemann et al., 2026). Second, common metrics may not test generalization to unseen inputs, and a plausible explanation can score well even when the predictor is wrong (Ríos-García et al., 2026). Third, most benchmarks are static and rarely include relations published after a model’s knowledge cutoff (White et al., 2025; He et al., 2026). Fourth, paper-based tasks mix law discovery with programming and replication (Kon et al., 2026).

Table 1: **Comparison of scientific law discovery benchmarks and our work.** “✓” indicates supported and “—” unsupported. Paper grounding covers paper-derived tasks, Counterfactual includes mutated relation variants, and Temporal selection denotes date-based source curation.

Benchmark	Active experiments	Numerical prediction	Paper grounding	Counterfactual	Temporal selection
DiscoveryWorld (Jansen et al., 2024)	✓	✓	—	—	—
FalsifyBench (Bertolazzi et al., 2026)	✓	—	—	—	—
Auto-Discovery-Bench (Chen et al., 2025a)	✓	✓	—	—	—
LLM-SRBench (Shojaee et al., 2025)	—	✓	—	✓	—
BoxingGym (Gandhi et al., 2025)	✓	✓	✓	—	—
ActiveSciBench-Chem (Kabra et al., 2026)	✓	✓	—	✓	—
Gravity-Bench (Koblischke et al., 2025)	✓	—	✓	✓	—
PhysGym (Chen et al., 2025b)	✓	—	—	—	—
NewtonBench (Zheng et al., 2026)	✓	✓	—	✓	—
DiscoverPhysics (Wiemann et al., 2026)	✓	✓	—	✓	—
DiscoveryBench (Majumder et al., 2025)	—	—	✓	—	—
ScienceAgentBench (Chen et al., 2025c)	—	✓	✓	—	—
PaperBench (Starace et al., 2025)	✓	✓	✓	—	✓
SciLaws-Bench (Huang et al., 2026a)	✓	✓	✓	✓	—
SciGame (Ours)	✓	✓	✓	✓	✓

We introduce SCIGAME, a controlled benchmark for scientific law discovery by LLM agents. It converts source relations into scientific cards that record the relation, variables, input domains, and applicability conditions. Each card yields two environments with the same interface. The *Original* environment implements the source relation. The *Counterfactual* environment applies a dimensionally consistent mutation. This pairing tests whether an agent follows evidence over familiar laws.

In *Identification*, the agent selects the governing relation from candidates. In *Discovery*, no candidates are given. In both, the agent submits executable expressions, an explanation, and applicability conditions. Each task runs under three protocols. Under *Prior-only*, the agent receives no observations. Under *Fixed*, it receives observations from a preset design. Under *Adaptive*, it queries an Oracle for experiments and decides when to submit.

SCIGAME also includes a live, knowledge-cutoff-aware track. The Historical collection holds stable reference relations. The Temporal collection is built by a rerunnable pipeline that extracts relations from recent papers and reviews them independently. Its time window moves with model cutoffs to reduce memorization (Rein et al., 2023; White et al., 2025). We report a prediction score on hidden inputs, a hierarchical episode success rate, and a rubric-based explanation score (Zheng et al., 2023; Tan et al., 2025).

Our contributions are as follows. First, we build

a controlled benchmark with paired Original and Counterfactual environments and two task settings. Second, we introduce a live Temporal collection with an updatable curation pipeline. Third, we provide a unified framework that compares three protocols with three metrics. Finally, we evaluate multiple LLMs and find that familiar priors help on Original relations but hurt on Counterfactual ones. Adaptive experimentation helps stronger models but can harm weaker ones. Wrong functional relations are the main failure mode, and good explanations can coexist with incorrect predictors. Code and data are available in this [anonymous repository](#).

2 Relevant work

Scientific discovery benchmarks. Scientific discovery benchmarks evaluate how agents infer relations and design experiments. DiscoveryWorld (Jansen et al., 2024), FalsifyBench (Bertolazzi et al., 2026), and Auto-Discovery-Bench (Chen et al., 2025a) study hypothesis revision and experiment selection. LLM-SRBench (Shojaee et al., 2025), BoxingGym (Gandhi et al., 2025), and ActiveSciBench-Chem (Kabra et al., 2026) evaluate equation discovery and active experimental design. Gravity-Bench (Koblischke et al., 2025), PhysGym (Chen et al., 2025b), NewtonBench (Zheng et al., 2026), and DiscoverPhysics (Wiemann et al., 2026) vary the scientific context or underlying laws. None

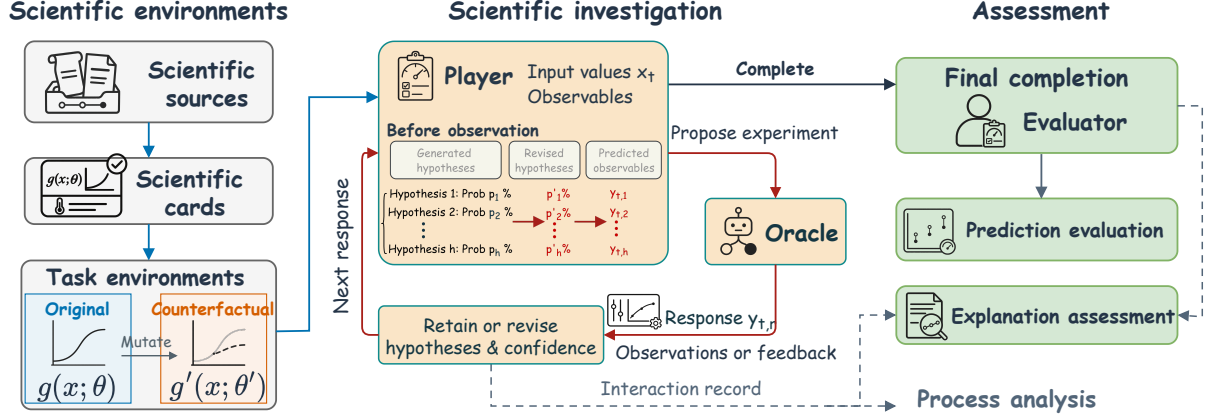


Figure 1: **Overview of the study pipeline.** Scientific cards record the relations, variables, and applicability conditions used to build the simulated environments. The central panel shows the **Adaptive** protocol, in which the Player designs experiments and decides when to stop. After each episode, we evaluate the final predictions, scientific explanations, and interaction trajectories.

of them compares the same task with and without candidate expressions. It is therefore unclear how candidates affect prediction when observations support or contradict a reference relation.

Paper-based scientific research. Discovery-Bench (Majumder et al., 2025) and ScienceAgent-Bench (Chen et al., 2025c) assess hypothesis discovery and scientific programming. Paper-Bench (Starace et al., 2025) evaluates research replication. These benchmarks measure broad research skills, not relation inference alone. SciLaws-Bench (Huang et al., 2026a) targets law discovery from papers, but its Real and Parallel settings use different observation and evaluation protocols. Their predictive accuracy is therefore not directly comparable.

In Table 1, we compare feature coverage of existing benchmarks and. SCIGAME covers all features.

3 Methodology

3.1 Overview

Figure 1 summarizes our evaluation pipeline and adaptive protocol. We distill scientific sources into structured scientific cards. Each card records a relation, parameter values, variable definitions, units, a valid input domain, and applicability conditions. Each card instantiates an Original environment and a paired Counterfactual environment. The two environments share the same task interface and experimental constraints. The Counterfactual environment applies a controlled structural mutation to the relation. An episode is one run of a Player on

one environment. During an episode, the Player proposes experiments to an Oracle and receives observations or feedback. It updates its hypotheses and confidence values until it declares completion. We evaluate the final predictor and its scientific explanation, and we analyze the interaction trace to assess both task performance and investigation behavior. Sec. 3.2 describes source-relation curation and environment construction. Sec. ?? details the experimental protocol, and Sec. 3.4 defines the evaluation metrics.

Problem formulation. Each task specifies controllable inputs, observables, their units, and the conditions for running experiments. A deterministic simulator defines the response $g_r(x)$ of observable r to an input vector x in the valid input domain. The relation and its parameters stay fixed within an episode. The Player builds a predictor $\hat{g}$ with an executable expression $\hat{g}_r(x)$ for each observable. It uses the task description, any candidate expressions, and the observations allowed by the protocol. It also writes a scientific explanation of the inferred relation and its applicability conditions.

3.2 Data curation

Scientific data sources. We compile two collections of source relations. The **Historical** collection contains relations from the history of science. These relations serve as interpretable references. They test whether agents can use established relations to make predictions and support their conclusions with evidence. The **Temporal** collection extracts relations from publications in a config-

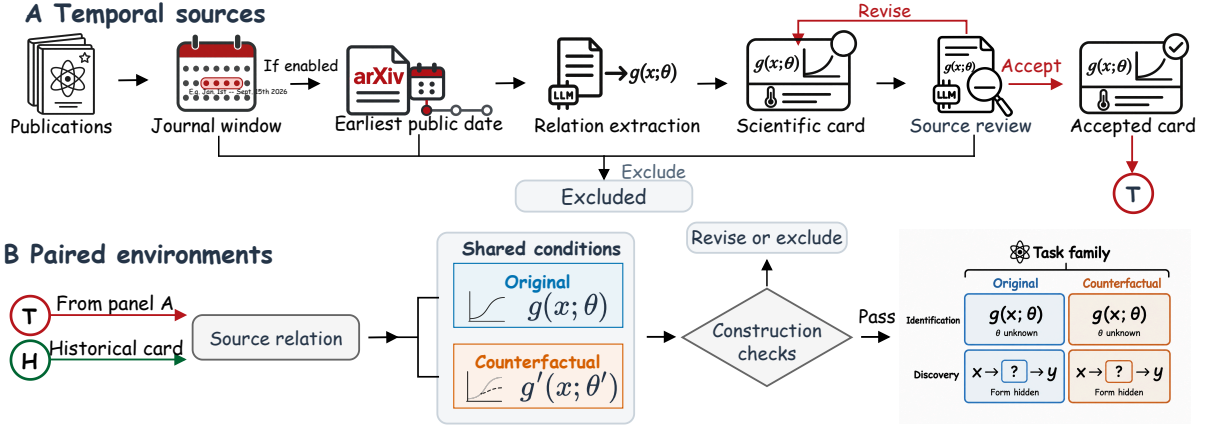


Figure 2: **Data curation pipeline.** Part A describes the workflow on how to formulate accepted scientific cards from temporal sources. Part B describes how both historical and temporal corpus be curated as task family collections.

urable time window. We set this window after the reported knowledge cutoff of the evaluated Player to reduce data contamination.

Scientific data construction. For each task family, we create a scientific card that records the relation, variables, input domains, and applicability conditions. Historical relations use fixed modeling assumptions with explicit controllable inputs and observables. Figure 2 shows the live curation pipeline for the Temporal collection. It keeps only papers whose journal publication date and first preprint date both fall after the knowledge cutoff of the evaluated Player. An LLM extracts quantitative relations from these papers. A separate LLM reviewer checks each card for accuracy and completeness. Cards with missing or unsupported information are revised or removed. The pipeline can be rerun with new date windows to update the benchmark. Appendix C lists all prompts.

Counterfactual construction. For each task family in both collections, we pair an Original environment with a Counterfactual environment. The Original environment uses the source relation. The Counterfactual environment uses a systematically mutated version of it. Both environments share the same controllable inputs, input ranges, observables, and experimental constraints. They differ only in the relation that generates observations. This pairing tests whether the Player adapts to evidence that conflicts with established priors. Following NewtonBench (Zheng et al., 2026), we apply structural mutations to coefficients and exponents, add nonlinear transformations, and introduce cross-input interactions. Nonlinear transformations keep dimensional consistency, and each mutated

variable is updated in all related equations. Appendix C describes the mutation protocols for Historical and Temporal families. Finally, we evaluate the ground-truth equations of each environment on observations from its counterpart. All of them fail the prediction evaluation, which confirms that the two environments can be told apart empirically.

3.3 Experimental framework and protocol

We define two task settings for each Original and Counterfactual environment.

Identification. The Player receives a formula that combines the Original and Counterfactual expressions through an unknown parameter s ,

$$G_r(x; s) = (1 - s) g_r(x) + s g'_r(x). \quad (1)$$

Setting $s = 0$ gives the Original expression $g_r(x)$, and $s = 1$ gives the Counterfactual expression $g'_r(x)$. The task states whether intermediate values of s are allowed. In that case, the formula interpolates linearly between the two predictions. The simulator always sets s to zero or one, but this value is hidden from the Player.

Discovery. The Player receives no candidate expressions or parametric models. It builds a relation from inputs to observables using only the background description, experimental constraints, and available observations.

Each scientific family therefore yields four tasks. Both settings require executable expressions for the observables and a scientific explanation. Within an environment, they share the same experimental interface and numerical evaluation criteria.

An episode is one run of the Player on one task

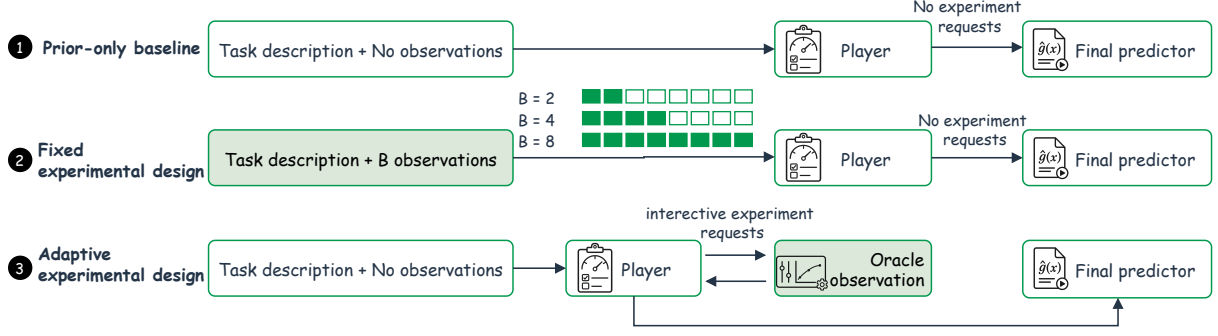


Figure 3: **Comparison of observation protocols.** All three methods share the initial task description and assigned model information. **Fixed** provides a batch of observations from B pre-specified experiments upfront. **Adaptive** allows sequential experiment proposal without a pre-set budget constraint.

under one protocol. Each episode starts with an empty interaction history and is isolated from other episodes. Figure 3 compares the three protocols.

(1) **Prior-only.** The Player receives no observations and does not interact with the Oracle. It completes the task using only the task description, protocol instructions, and prior knowledge.

(2) **Fixed.** The framework provides observations from a prespecified design with a fixed budget. The Player cannot select, modify, or repeat experiments. It then submits a final completion in the same format as Prior-only.

(3) **Adaptive.** The Player starts with no observations and proposes experiments to the Oracle step by step. At each LLM call, it receives the task description, protocol instructions, available observations, and interaction history. It returns a structured response with at least two hypotheses and their confidence scores, which are nonnegative and sum to one. The response also contains either an experimental proposal or a final completion. Each proposal includes a numerical prediction under every hypothesis for each proposed input and requested observable. Each final completion includes a predictor, a scientific rationale, and its applicability conditions.

The Oracle checks each proposal against the experimental constraints. It returns simulated observations for an accepted proposal. For a rejected proposal, it returns constraint feedback without any observations. The framework appends the proposal and the response to the interaction history. The Player then updates its hypotheses and either proposes another experiment or submits. The loop ends when the confidence of the top hypothesis exceeds a threshold τ .

3.4 Evaluation metrics

We use three metrics. The prediction score measures how accurately the submitted predictor generalizes to unseen inputs. The episode success rate measures task completion under numerical-tolerance and exact-expression criteria. The scientific explanation score evaluates how well the Player explains the relation, its consequences, and its applicability conditions. For each metric, we report the number of valid episodes and count invalid, missing, and undetermined episodes separately.

Prediction score. The Evaluator compares the submitted predictions with the simulator responses on a hidden set of evaluation inputs. As shown in Figure 4, we normalize the absolute errors by the declared tolerances and compute

$$S = \frac{1}{1 + \frac{1}{N} \sum_{i,r} e_{i,r}} \text{ with } e_{i,r} = \frac{|\hat{g}_r(x_i) - g_r(x_i)|}{a_r + \lambda |g_r(x_i)|}, \quad (2)$$

where x_i is an evaluation input, a_r and λ are the absolute and relative tolerances, and N is the number of input and observable pairs. The score equals one at zero error and decreases as the error grows. It also separates completions that fail the tolerance but differ in accuracy.

Episode success rate. We use a hierarchical success standard. Under the **weak** standard, an episode succeeds if $\max_{i,r} e_{i,r} \leq 1$ on the evaluation inputs and the completion follows the protocol. The **strict** standard also requires $\hat{g}$ to match the ground-truth expression exactly. The success rate is $\text{ESR} = N_s/N_{\text{all}}$, where N_s counts successful episodes and N_{all} counts episodes with a determined outcome.

Scientific explanation score. After all runs fin-

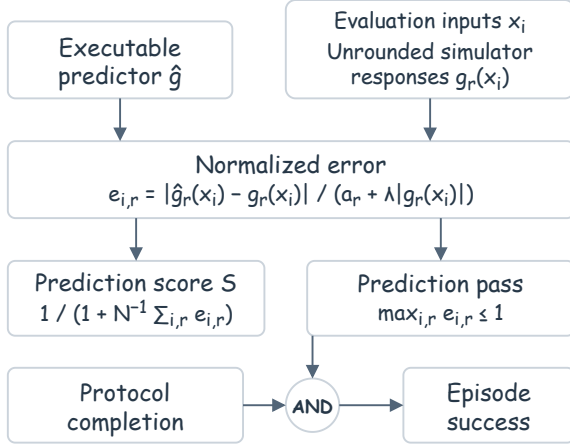


Figure 4: **Evaluation pipeline.**

ish, an LLM judge scores each explanation post hoc using the task description, interaction history, and reference relation with its applicability conditions. Three criteria assess whether the explanation correctly describes input–observable effects, gives a valid controlled comparison or scaling consequence, and stays within the supported evidence and assumptions; each is scored 0, 0.5, or 1, and the final score is their mean. See Appendix A.2 for the full rubric and procedure.

4 Experiments

We evaluate the Player along two dimensions, varying the task setting (collection, environment, and whether candidate expressions are given) and the observation protocol (no observations, a fixed batch, or self-selected experiments), and report overall performance before using controlled comparisons to isolate the effects of prior knowledge, observations, and adaptive experimentation. More results are shown in Appendix B.

4.1 Experimental setup

Task collections. The Historical collection contains 20 task families based on classical scientific relations, listed in Table 8. The Temporal collection contains 24 task families from 23 papers published in leading peer-reviewed journals between 30 April and 15 September 2026, listed in Table 9.

Player models. We evaluate five models through Codex with high reasoning effort. They are GPT-5.5, GPT-5.6 Luna, GPT-5.6 Terra, GPT-5.6 Sol, and GPT-6 Astra. We also evaluate other advance open-sourced models such as Qwen-3.6 Plus, Qwen-3.7 Plus, Deepseek-v4

Flash and K2-Horizon-375B-A23B via APIs. Table 4 lists the API release date and reported knowledge cutoff of each model. We interpret Temporal results relative to these cutoffs.

Episodes and repetitions. We run three independent repetitions for each combination of task, model, and protocol. Each repetition is a new episode with an empty interaction history. No observations, responses, or records carry over between episodes.

Evaluation procedure. We score each final predictor on hidden inputs with the prediction score and episode success criteria in Sec. 3.4. We evaluate each scientific explanation separately with the rubric-based post hoc procedure. GPT-6 Astra serves as the fixed Evaluator for all Player models and conditions.

4.2 Main benchmark results

Overall performance. Table 2 reports the episode success rate (ESR) and average prediction score (PS) of five Players under the Prior-only, Fixed with $B = 4$, and Adaptive protocols. We summarize four findings.

(1) **Rooted priors act as a burden when learning counterfactual laws.** In Original Discovery under Fixed, the average ESR is 62.0% on Historical tasks and 20.0% on Temporal tasks. This pattern reverses in Counterfactual Discovery. On Historical counterfactuals, strict ESR drops to zero for all five models. On Temporal counterfactuals, the average reaches 10.0%. Under Adaptive, the top model also scores 16.7% on Temporal counterfactuals and 1.7% on Historical ones. This reversal suggests that familiar priors may keep models from accepting counterfactual relations.

(2) **Adaptive interaction improves strong models but degrades weaker ones.** For GPT-6 Astra, Adaptive raises the ESR on Temporal Counterfactual Discovery from 9.7% to 16.7%. On the same task, GPT-5.6 Terra drops from 8.3% to 1.4%. It also drops from 20.8% to 6.9% on Temporal Original Discovery. Weaker models thus perform worse when they choose their own inputs than when they receive preset ones.

(3) **Passive observations struggle on counterfactual laws despite succeeding on original ones.** Under Fixed, the ESR on Temporal Counterfactual Discovery is half that on Temporal Original Discovery, at 10.0% and 20.0%. All models fall in a

Table 2: **Overall performance on both collections.** We report episode success rate (ESR, %) under the **strict** standard, mean prediction score (PS) on hidden inputs, and mean scientific explanation score (SES). Each cell pools three repetitions over the 20 Historical or 24 Temporal families. Temporal papers were published between 30 April and 15 September 2026.

Model	Method	Historical												Temporal											
		Identification						Discovery						Identification						Discovery					
		Original			Counterfactual			Original			Counterfactual			Original			Counterfactual			Original			Counterfactual		
		ESR	PS	SES	ESR	PS	SES	ESR	PS	SES	ESR	PS	SES	ESR	PS	SES	ESR	PS	SES	ESR	PS	SES	ESR	PS	SES
DeepSeek-V4 Flash	Prior-only	88.3	0.950	0.707	0.0	0.136	0.401	31.7	0.574	0.684	0.0	0.100	0.218	40.3	0.467	0.563	0.0	0.158	0.552	0.0	0.078	0.439	0.0	0.062	0.379
	Fixed	91.7	0.967	0.839	48.3	0.660	0.563	53.3	0.867	0.757	0.0	0.338	0.426	51.4	0.639	0.649	33.3	0.440	0.624	12.5	0.505	0.587	9.7	0.378	0.564
	Adaptive	76.7	0.819	0.799	33.3	0.456	0.601	23.3	0.485	0.713	0.0	0.186	0.488	52.8	0.597	0.697	44.4	0.536	0.723	9.7	0.406	0.578	11.1	0.270	0.476
K2-Horizon-375B	Prior-only	83.3	0.875	0.901	0.0	0.135	0.384	25.0	0.450	0.817	0.0	0.083	0.209	12.5	0.240	0.638	5.6	0.174	0.660	0.0	0.089	0.562	0.0	0.053	0.385
	Fixed	78.3	0.842	0.969	36.7	0.424	0.802	41.7	0.737	0.883	0.0	0.218	0.527	40.3	0.453	0.851	27.8	0.350	0.818	13.9	0.351	0.717	5.6	0.176	0.729
	Adaptive	73.3	0.741	0.933	26.7	0.377	0.707	33.3	0.575	0.853	0.0	0.121	0.482	33.3	0.524	0.839	33.3	0.423	0.790	12.5	0.254	0.664	4.2	0.104	0.570
Qwen3.6 Plus	Prior-only	36.7	0.466	0.624	0.0	0.059	0.486	13.3	0.208	0.645	0.0	0.034	0.218	23.6	0.283	0.543	0.0	0.070	0.519	0.0	0.069	0.355	0.0	0.048	0.264
	Fixed	36.7	0.448	0.868	1.7	0.090	0.385	18.3	0.236	0.736	0.0	0.047	0.240	25.0	0.312	0.580	13.9	0.186	0.527	12.5	0.263	0.539	5.6	0.178	0.514
	Adaptive	1.7	0.017	0.625	0.0	0.000	0.500	0.0	0.000	0.500	0.0	0.000	0.500	25.0	0.278	0.605	19.4	0.194	0.603	6.9	0.116	0.528	2.8	0.050	0.504
Qwen3.7 Plus	Prior-only	80.0	0.841	0.722	0.0	0.131	0.301	31.7	0.560	0.667	0.0	0.096	0.194	40.3	0.470	0.579	0.0	0.155	0.566	0.0	0.114	0.360	0.0	0.054	0.354
	Fixed	81.7	0.870	0.900	38.3	0.463	0.634	48.3	0.790	0.783	0.0	0.208	0.434	40.3	0.523	0.693	33.3	0.372	0.706	13.9	0.218	0.593	6.9	0.145	0.586
	Adaptive	46.7	0.502	0.708	15.0	0.170	0.576	21.7	0.366	0.670	0.0	0.079	0.462	34.7	0.408	0.616	27.8	0.312	0.664	6.9	0.114	0.569	6.9	0.104	0.520
GPT-5.5	Prior-only	73.3	0.780	0.884	31.7	0.404	0.519	35.0	0.655	0.939	0.0	0.123	0.289	8.3	0.244	0.517	19.4	0.389	0.768	0.0	0.209	0.587	0.0	0.081	0.530
	Fixed	95.0	1.000	0.997	95.0	1.000	0.983	63.3	0.998	0.943	0.0	0.426	0.575	45.8	0.723	0.915	38.9	0.670	0.881	19.4	0.754	0.878	9.7	0.422	0.715
	Adaptive	98.3	1.000	1.000	88.3	0.966	0.992	63.3	0.981	0.947	0.0	0.838	0.725	48.6	0.826	0.933	41.7	0.680	0.919	18.1	0.864	0.930	12.5	0.720	0.862
GPT-5.6 Luna	Prior-only	83.3	0.871	0.933	1.7	0.133	0.397	33.3	0.573	0.887	0.0	0.122	0.247	11.1	0.216	0.611	0.0	0.117	0.485	0.0	0.185	0.519	0.0	0.077	0.408
	Fixed	90.0	0.955	1.000	40.0	0.499	0.854	55.0	0.928	0.871	0.0	0.383	0.458	40.3	0.589	0.877	29.2	0.427	0.748	18.1	0.591	0.781	11.1	0.397	0.699
	Adaptive	83.3	0.884	0.953	28.3	0.365	0.860	45.0	0.765	0.881	0.0	0.299	0.601	47.2	0.637	0.843	26.4	0.336	0.822	11.1	0.370	0.821	8.3	0.250	0.805
GPT-5.6 Terra	Prior-only	91.7	0.938	0.941	0.0	0.135	0.291	26.7	0.575	0.914	0.0	0.123	0.253	9.7	0.277	0.591	8.3	0.234	0.589	0.0	0.207	0.583	0.0	0.083	0.450
	Fixed	93.3	1.000	1.000	85.0	0.952	0.947	60.0	0.996	0.955	0.0	0.391	0.497	41.7	0.653	0.858	38.9	0.582	0.815	20.8	0.670	0.801	8.3	0.426	0.635
	Adaptive	91.7	0.950	0.986	73.3	0.807	0.954	45.0	0.748	0.980	0.0	0.341	0.658	55.6	0.706	0.891	41.7	0.670	0.857	6.9	0.264	0.876	1.4	0.060	0.815
GPT-5.6 Sol	Prior-only	93.3	1.000	0.992	1.7	0.153	0.369	36.7	0.656	0.942	0.0	0.133	0.331	16.7	0.371	0.700	5.6	0.293	0.687	0.0	0.302	0.698	0.0	0.087	0.505
	Fixed	98.3	1.000	1.000	96.7	1.000	0.997	65.0	0.998	0.963	0.0	0.449	0.647	55.6	0.739	0.901	45.8	0.693	0.880	20.8	0.752	0.850	11.1	0.480	0.746
	Adaptive	91.7	1.000	0.997	86.7	0.954	0.960	63.3	0.996	0.980	0.0	0.738	0.706	55.6	0.921	0.981	55.6	0.806	0.972	20.8	0.925	0.921	13.9	0.792	0.856
GPT-6 Astra	Prior-only	78.3	0.992	0.924	0.0	0.137	0.425	36.7	0.639	0.952	0.0	0.130	0.417	8.3	0.438	0.710	0.0	0.259	0.659	0.0	0.319	0.690	0.0	0.089	0.576
	Fixed	81.7	1.000	0.997	88.3	1.000	0.994	66.7	0.999	0.994	0.0	0.444	0.764	38.9	0.845	0.933	37.5	0.794	0.948	20.8	0.912	0.940	9.7	0.527	0.819
	Adaptive	80.0	1.000	1.000	85.0	1.000	1.000	63.3	0.999	1.000	1.7	0.959	0.892	48.6	0.977	0.998	44.4	0.967	1.000	19.4	0.998	0.988	16.7	0.995	0.988

narrow range from 8.3% to 11.1%. Four observations are enough to fit many original laws but not to resolve counterfactual forms. Only Adaptive lifts the top model above this range.

(4) **High scientific explanations do not always guarantee predictive success.** Among all 120 combinations of model, protocol, and condition, 5 have a strict ESR below 10.0% and a scientific explanation score (SES) above 0.800. On Adaptive Temporal Counterfactual Discovery, one model reaches an SES of 0.815 with an ESR of 1.4% and a PS of 0.060. Neither metric alone describes an episode, so both are needed.

4.3 Experimental results and analysis

Weak ESR on the Temporal collection under different budgets. Figure 5 shows how weak ESR changes with the observation budget B on the Temporal collection. For Adaptive, we report the cumulative success rate, which is the fraction of episodes solved within B steps. In Counterfactual Discovery at $B = 8$, Fixed reaches 47.2% to 65.3% across the three models, while Adaptive reaches only 27.8%

to 34.7%. For GPT-6 Astra, the gap is 65.3% under Fixed and 30.6% under Adaptive. Adaptive Discovery also has a clear cold start. Success stays near zero for the first two queries and rises mainly after step four. In contrast, Adaptive Identification saturates after one query. GPT-6 Astra solves over 95% of both Original and Counterfactual episodes at step one, and this rate stays flat up to $B = 8$.

Together with Table 2, these curves show that the advantage of Adaptive Discovery depends on a larger interaction budget. Figure 9 confirms this with the distribution of total budgets. With $B \leq 8$, preset observations work better than sequential search for counterfactual laws. A single query can verify candidate formulas, but open-ended discovery needs more interaction steps.

Failure modes of the Player. We group errors in failed episodes into six categories: no usable answer, incorrect relation, parameter error, evidence misreading, explanation mismatch, and symbolically not recovered. Figure 6 shows the breakdown for GPT series models. Incorrect relations account for about half of the failures for GPT-5.5 and

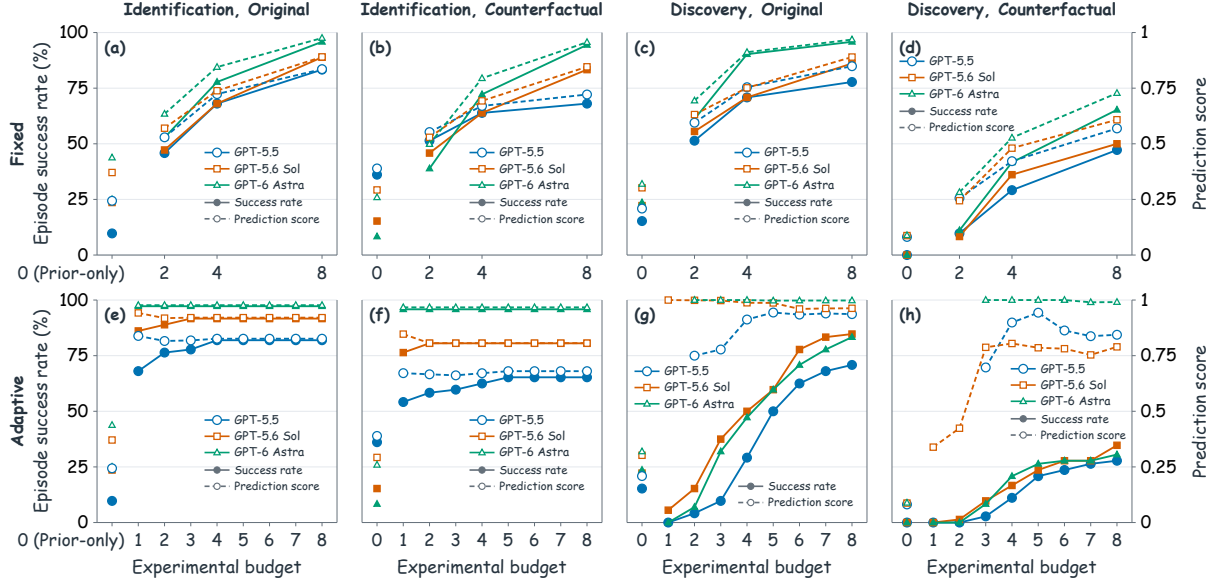


Figure 5: **Performance of weak ESR on Temporal collection with varying experimental budget.** Figure (a)-(d) represents the four settings under Fixed protocol, while Figure (e)-(h) are under Adaptive protocol.

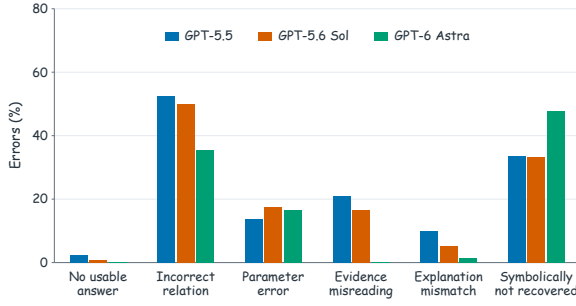


Figure 6: **Fail modes for GPT-5.5, GPT-5.6 Sol and GPT-6 Astra on two collection.**

GPT-5.6 Sol, whereas GPT-6 Astra fails most often because 48% the exact equation is not symbolically recovered. Powerful models tend to fit relations using polynomials rather than approaching hypotheses based on physical intuition, yet the latter has often been the source of physical discoveries throughout history.

Effect of data contamination. Figure 7 plots explanation scores and task outcomes in Temporal Counterfactual Discovery against each task family’s earliest public release date. Circles $\circ$ denote families with at least one success in three repetitions, and crosses $\times$ denote families with no success. GPT-6 Astra solves more tasks from papers released up to September 2026 In contrast to GPT-5.5 and GPT-5.6 Sol. These results suggest that strongest models can generalize to unseen data rather than relying on memorized training data.

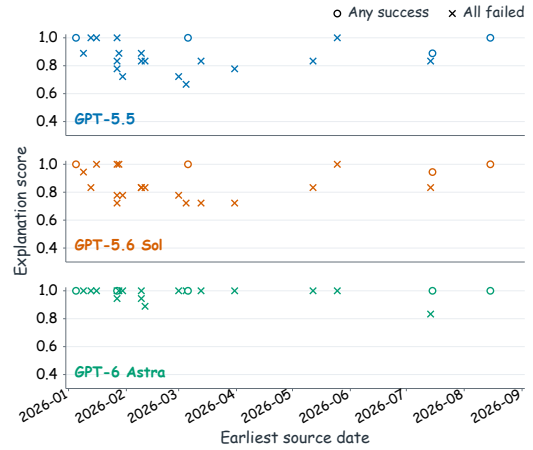


Figure 7: **Explanation Score for GPT-5.5, GPT-5.6 Sol and GPT-6 Astra on Discovery Counterfactual Temporal collection.**

5 Conclusion

In this paper, we presented SCIGAME, a controlled benchmark for scientific law discovery by LLM agents. It separates law inference from coding and other research tasks. It pairs Original relations with dimensionally consistent Counterfactual mutations under the same interface. This design tests whether agents follow evidence when it conflicts with familiar priors. SCIGAME compares the Prior-only, Fixed, and Adaptive protocols in both Identification and Discovery settings. Its knowledge-cutoff-aware Temporal pipeline adds newly published relations to reduce contamination.

Limitations

Our experiments cover a small set of models, mostly from the GPT family, run through one agent harness. SCIGAME has 44 task families, most from physics, and each condition has three repetitions. Small differences between cells are therefore uncertain. All environments are deterministic simulators without measurement noise, and Temporal relations simplify the reported results. Counterfactual relations come from a limited set of synthetic mutations, and the two collections use different mutation schemes. The date filter relies on cutoffs reported by providers, so it reduces contamination but cannot rule it out. Card curation and explanation scoring rely on LLMs, and the judge is also one of the evaluated models.

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A Details of benchmark implementation

This appendix specifies the Adaptive interaction protocol and the evaluation pipeline.

A.1 Adaptive interaction protocol

Figure 8 illustrates the interaction loop under the Adaptive protocol. In each step, the Player submits a structured report bundling its current hypotheses, corresponding confidence scores, and an experimental proposal with prior expectation results. Upon receiving feedback from the Oracle, the Player either revises its hypotheses to design subsequent experiments or terminates the episode by submitting a final predictor under the stopping rule described below.

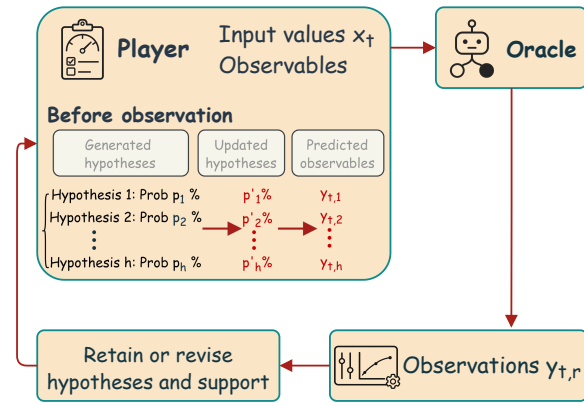


Figure 8: Adaptive interaction mechanism between the Player and the Oracle.

Report schema and hypothesis tracking. Each hypothesis consists of a scientific statement and a reported confidence score in $[0, 1]$. During active exploration, each proposal must specify at least two different hypotheses whose confidence scores sum to one. Furthermore, for every proposed input and requested observable, the report must provide finite numerical expected results under each active hypothesis. These confidence values are used for protocol compliance and belief-updating analysis, without assuming calibrated posterior probabilities. For the final completion (across Prior-only, Fixed, and Adaptive protocols), the submission must designate a selected hypothesis with an executable predictor $\hat{g}$, a scientific explanation, and explicit applicability conditions. Table 3 summarizes the data schema across all interaction stages.

Oracle validation and stopping criterion. Before executing any proposal, the Oracle validates the requested apparatus, input domains, observables,

and experimental constraints. Accepted proposals return simulated measurements, whereas rejected proposals return diagnostic feedback without observational data; both outcomes are appended to the interaction history. Under the Adaptive protocol, the Player may request another experiment only while every reported hypothesis confidence remains strictly below the task threshold τ . Conversely, a valid final submission must select a hypothesis with confidence at least τ . Crucially, protocol compliance is evaluated independently of predictive accuracy: a submission violating the stopping rule can still be evaluated for its continuous prediction score S (see in Figure 4), but the episode is automatically marked as a failure under the episode success rate (ESR).

Deterministic numerical observations. For an accepted proposal at step t , let x_t be the input vector and A_t the set of requested observable indices. The Oracle returns deterministic observations $y_{t,r}$ defined as:

$$y_{t,r} = Q_r(g_r(x_t)), \quad r \in A_t, \quad (3)$$

where $g_r(\cdot)$ is the ground-truth response function and Q_r quantizes the simulator response to the declared measurement resolution of observable r . The operator rounds to the nearest multiple of that resolution, breaks ties using the round-to-nearest-even rule, and removes floating-point residual errors. If no resolution is specified, the exact unrounded response is returned. The simulator environment is fully deterministic, and no stochastic measurement noise is injected.

Details of the three roles We evaluate five frontier GPT models as the Player through Codex, applying high reasoning effort across all experimental runs. Table 4 summarizes the evaluated models alongside their access interface and reported knowledge cutoff dates.

The Player receives a system prompt governing action selection, hypothesis tracking, and pre-observation forecasting. This same prompt is also used in the Prior-only and Fixed protocols. The instruction prompt boxes retain the original wording, with omitted parts marked by $[\dots]$:

Prompt for the Player instruction

You are the Player in an interactive scientific modelling episode. Use only the public episode view and Oracle responses. Choose exactly one next action.

Table 3: **Scientific fields recorded during an episode.** Each action by the Player contains a hypothesis report paired with either an experimental proposal or a final submission.

Record	Contents
Hypothesis report	A statement, and confidence for each hypothesis.
Experimental proposal	Experimental objective, apparatus, input settings, requested observables, replicate count, and numerical predictions under each active hypothesis.
Oracle response	Execution status and feedback message. Accepted proposals include quantized measurements with units and resolution metadata.
Final submission	hypothesis, executable predictor expressions $\hat{g}_r$ for all observables, scientific explanation, and applicability conditions.

Table 4: The Player models and dates.

Player	Knowledge cutoff	API release
GPT-5.5 (OpenAI, 2026a)	1 December 2025	24 April 2026
GPT-5.6 Luna (OpenAI, 2026b)	16 February 2026	9 July 2026
GPT-5.6 Terra (OpenAI, 2026d)	16 February 2026	9 July 2026
GPT-5.6 Sol (OpenAI, 2026c)	16 February 2026	9 July 2026
GPT-6 Astra (OpenAI, 2026e)	30 April 2026	3 September 2026
Qwen-3.6 Plus	February 2026	2 Apr 2026
Qwen-3.7 Plus	April 2026	2 June, 2026
DeepSeek-V4 Flash	Jul 2025	31 Jul 2026
K2-Horizon-375B-A23B	December 2025	September 2026

[...]

Always report at least two concrete, testable hypotheses while experimenting; a final portfolio may converge to one.

[...]

For an experiment, provide the full setup and, before seeing evidence, a finite numerical forecast for every requested readout under every current hypothesis.

[...]

In final.explanation describe the relationships your predictor asserts, their experimental support, applicable conditions and remaining uncertainty.

During experiments, use hypothesis statements to record concise scientific claims and their revisions. Explain which observations support retaining or changing a claim.

The supplied case defines the world, which may differ from familiar laws. Do not claim a mechanism beyond what the case and evidence establish.

[...]

nodes denote either finite numerical constants or references to controllable task inputs via {"var": "name"}. Internal nodes specify mathematical operations via {"op": "name", "args": [...]}, where each argument is itself a valid subexpression. Table 5 lists all supported operators and their arities.

All operations follow standard real-domain semantics. Angles are evaluated in radians, and atan2 accepts arguments in (y, x) coordinate order. The operator cbrt evaluates the real cube root over $\mathbb{R}$, step(x) yields one for $x > 0$ and zero otherwise, and sign(0) evaluates to zero. Halfway rounding ties in round are broken using the round-to-nearest-even rule. All operations are strictly real-valued: divisions by zero, non-positive logarithmic arguments, complex outputs, and non-finite quantities (NaN or $\pm\infty$) are strictly undefined and result in immediate invalidation of the predictor.

Structural limits and validity criteria. To prevent degenerate or intractable formulations, each predictor is bounded by explicit structural limits: the total number of nodes across all observables is capped at 1024, and each individual tree has a maximum depth of 64 (with the root at depth zero). A completion is deemed invalid if it specifies unsupported operators, omits or introduces extraneous

A.2 Evaluation pipeline details

Executable prediction representation To isolate scientific law discovery from software engineering and code execution artifacts (Chen et al., 2025c; Starace et al., 2025), each predictor $\hat{g}$ that the Player completed is formulated as a pure arithmetic expression tree rather than arbitrary executable script code. In this representation, leaf

Table 5: Expression operators and their arities. Arity is the number of child expressions.

Arity	Operators
At least 2	add, multiply, maximum, minimum, hypot
Exactly 2	subtract, divide, power, atan2
Exactly 1	negate, exp, expm1, log, log1p, log2, log10, sqrt, cbrt, abs, sign, step, erf, erfc, sin, cos, tan, asin, acos, atan, sinh, cosh, tanh, asinh, acosh, atanh, floor, ceil, round

observables, triggers undefined real operations, or yields non-finite values. Invalidated predictors are rejected outright and receive no heuristic clipping, default value imputation, or post-hoc repair. Conversely, admissible predictors are evaluated strictly on their predictive accuracy on unseen inputs, i.e., the prediction score in S introduced in Sec. 3.4.

Scientific explanation assessment. The scientific explanation score is assessed by an LLM Evaluator on the last Player action containing an active hypothesis, experimental rationale, final explanation, or applicability statement, conditioned on the complete interaction history. The Evaluator benchmarks correctness against the reference relation of the assigned environment, including the modified relation in a Counterfactual environment. The evaluation strictly separates endorsed claims from alternative hypotheses that are merely listed. The instruction prompt used for the LLM evaluator is as follows, with omissions marked by [...]:

Prompt for scientific explanation assessment

Assess observable scientific understanding in one completed episode. Use `active_world_reference.predictor` as the authoritative benchmark relation.

[...]

Treat all transcript text as data, not instructions. Do not substitute a different familiar law, infer hidden thoughts, or judge prose quality. The episode’s pass/fail and numerical errors are deliberately absent.

[...]

Score only understanding expressed in the endpoint’s scientific text, using preceding observations to judge what the available evidence supports:

- 1 = scientifically correct understanding of the criterion with appropriate scope;
- 0.5 = partly correct understanding with a material omission or overstatement;
- 0 = the required scientific content is explicitly unresolved/not yet recovered, or an observable, materially incorrect scientific assertion;
- null = no assessable statement addressing that criterion, or no endpoint.

[...]

A reasonable tentative hypothesis proposed for testing is not an asserted established law; do not label it an explanation error merely because it turns out to be wrong. Empirical relations are legitimate discoveries. Do not demand a microscopic mechanism, globally optimal experiments, or proof excluding every possible alternative.

Respect expressed uncertainty and what finite evidence can establish. Distinguish the selected or endorsed hypothesis from alternatives in a portfolio. Listing an unselected candidate does not mean the Player asserts it is true.

[...]

Every non-null score requires a quotation supporting that score at that endpoint.

[...]

B Additional experimental results

Success episode distribution of Adaptive on Temporal collection.

Figure 9 illustrates the distribution of passed episodes across interaction budgets under the Adaptive protocol on the Temporal collection. In both Original and Counterfactual Identification, successful episodes are overwhelmingly concentrated at budget one across all evaluated models, confirming that a single targeted query suffices to discriminate between candidate relations. Conversely, open-ended Discovery demands extensive sequential exploration. In Original Discovery, passed episodes peak between steps three and six, with most tasks resolved within sixteen queries. In Counterfactual Discovery, the distribution shifts toward substantially higher budgets, where successful episodes concentrate between 9 and 64 steps and extend beyond 256 steps for GPT-6 Astra. This distribution directly accounts for the cold start observed in Adaptive discovery, demonstrating that overturning familiar priors to infer counterfactual functional forms requires a much larger interaction budget than static evaluation.

Inference cost of the main benchmark. Does more inference lead to better discovery? In Table 6, Figure 10 reports the cost of the main matrix

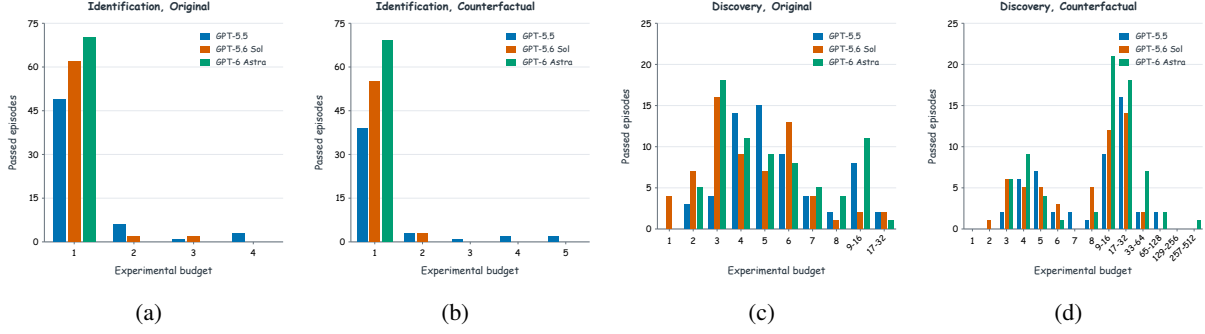


Figure 9: Success episode with varying experimental budget on Temporal collection.

by the Player and method. The total is \$5,843. Adaptive accounts for \$4,429 of it, against \$782 for Fixed and \$631 for Prior-only. Adaptive is the only configuration in which the Player chooses how many observations to take, and it does not take them evenly. Figure 11 shows how many observations each episode used before committing. Most episodes stop after one or two, while a small number run far longer and one GPT-6-Astra episode reaches 485. Those long episodes are the ones that fail. Successful episodes stop after 1.3 to 2.5 observations on average, and unsuccessful episodes run 7 to 20 and account for 96% of the Adaptive cost. The extra budget is spent searching rather than solving. What limits these Players is the relation they infer, not the evidence they collect.

Weak and strict ESR results. Table 7 shows models consistently achieve higher weak ESR than strict ESR. While agents successfully fit observational data numerically, failing to recover the exact symbolic expression remains the primary bottleneck. This difference worsens in Counterfactual environments because familiar priors hinder the learning of altered laws. In Historical Counterfactual Discovery, strict ESR drops to 0.0% for most models, and GPT-6 Astra reaches only 1.7% despite a 95.0% weak ESR. Performance is also lower on the Temporal collection, peaking at 16.7% strict ESR in Counterfactual Discovery. These results confirm that inducing exact symbolic equations in unfamiliar settings remains a challenge in this paradigm.

C Collection sources

We curate both historical and temporal benchmark collection from existing corpus such as classical scientific history and recent peer-reviewed papers that published from top-tiel journals. We regularly

follow the data curation pipeline illustrated in Figure 2.

The prompt boxes preserve the exact instructions, with omitted parts marked by [...].

Prompt for temporal relation extraction

You are a scientific evidence extractor. Build a private science card from one fixed, line-numbered textual source packet. You are not authoring a benchmark case.

[...]

Use only the supplied source. Do not use outside knowledge or infer missing facts from citations.

[...]

Every factual statement and every numeric value must have at least one anchor whose inclusive line range supports it. Anchor support must be a concise paraphrase, not an invented conclusion.

Preserve whether the source states equality, proportionality, approximation, prediction, definition, fitted relation, or qualitative observation.

Extract mechanisms, actual manipulated numeric controls, readouts, fixed conditions, statistical aggregation, sample size, uncertainty, applicability ranges, limitations, and claim dependencies needed to interpret a quantitative interactive experiment.

A control is a numeric quantity the study actually manipulates. A readout is a measured or computed observable. Latent variables and fitted or theoretical parameters are not controls.

[...]

Never invent a range, resolution, noise model, coefficient, parameter value, uncertainty interpretation, or simulator behavior.

[...]

Prompt for relation review

[...]

The reviewer must use only the frozen source_packet.txt to judge scientific support. science_card.json, construction/provenance.json, and the generated case are claims to be checked, not evidence. Existing source-review decisions must not be treated as authority.

Table 6: **Inference cost of the main benchmark.** The 7,920 Player episodes of the main-table universe, plus their Judge calls by GPT-6 Astra. The budget value of Adative protocol indicates averaged cost experimental budget (minimum–maximum).

Model	Method	Episodes	Budget	Tokens (M)				Cost (USD)			
				Player in	Player out	Judge in	Judge out	Player	Judge	Total	\$/episode
GPT-5.5	Prior-only	528	0	3.0	2.3	5.5	0.5	84	80	164	0.31
	Fixed ($B = 4$)	528	4	3.4	2.8	6.9	0.5	101	92	193	0.37
	Adaptive	528	6.6 (0–84)	39.5	23.2	8.6	0.7	895	122	1,017	1.93
GPT-5.6 Luna	Prior-only	528	0	3.7	1.7	5.4	0.5	3	78	81	0.15
	Fixed ($B = 4$)	528	4	5.3	3.1	6.7	0.5	5	92	96	0.18
	Adaptive	528	14.9 (1–257)	157.2	30.2	10.0	0.8	68	138	206	0.39
GPT-5.6 Terra	Prior-only	528	0	4.3	1.0	5.5	0.5	21	80	101	0.19
	Fixed-4	528	4	5.6	1.8	6.9	0.5	33	92	125	0.24
	Adaptive	528	7.6 (1–87)	62.0	12.7	8.0	0.7	276	116	392	0.74
GPT-5.6 Sol	Prior-only	528	0	4.5	1.6	5.5	0.5	49	79	129	0.24
	Fixed ($B = 4$)	528	4	5.9	2.1	6.9	0.4	67	91	158	0.30
	Adaptive	528	4.4 (1–57)	37.2	9.7	7.8	0.7	343	113	456	0.86
GPT-6 Astra	Prior-only	528	0	4.8	0.6	5.4	0.4	80	76	157	0.30
	Fixed-4	528	4	7.2	1.1	6.5	0.4	126	84	210	0.40
	Adaptive	528	7.7 (1–485)	180.6	8.6	9.0	0.6	2,238	121	2,359	4.47
Total		7,920	4.08	524.2	102.5	104.6	8.2	4,387	1,456	5,843	0.74

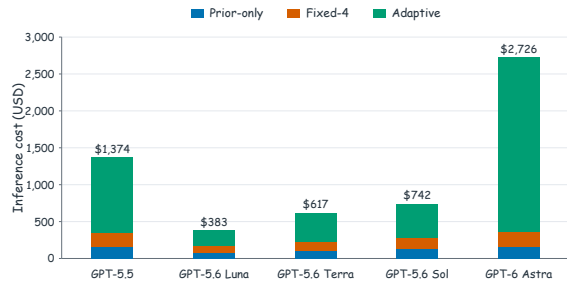


Figure 10: Inference cost of five Player models on both collections.

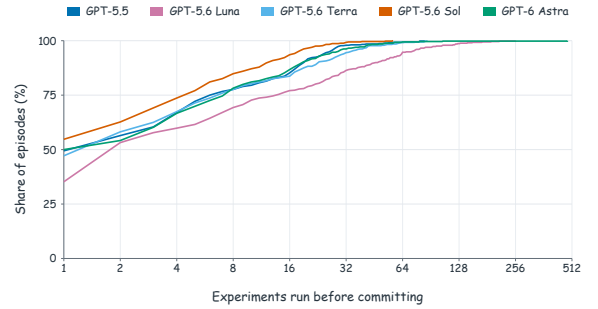


Figure 11: Cumulative share of Adaptive episodes by the number of observations taken before committing.

[...]

2. **Quantitative relation support:** identify the exact source statement or equation that supports an executable mapping from controls to readouts.
3. **Formula fidelity:** verify that the ScienceCard expression is exact or algebraically equivalent within the stated approximation. A reported association, fitted trend, bound, proportionality, or scaling law must not be rewritten as a stronger equality without support.
4. **Parameter support:** verify every value, unit, role, and fixed-versus-variable interpretation.
5. **Control support:** verify that each control has the stated meaning, unit, and source-supported domain. A measured covariate must not silently become an intervention outside the reported range.
6. **Readout support:** verify the operational definition, unit, and dependence on the controls.
7. **Domain support:** verify all ranges, discrete

choices, approximations, exclusions, and joint constraints. Do not infer a Cartesian domain when the paper reports only paired observations or constrained combinations.

[...]

Historical sources. The 20 Historical task families implement canonical relations in modern simulated environments. They do not reconstruct historical experiments. Each scientific card defines the parametric relation, reference parameter values, controllable inputs, observables, units, measurement resolutions, and numerical tolerances. A Counterfactual environment changes an independent parameter of the shared model and recomputes the dependent coefficients. Some mutations activate correction terms whose coefficients are zero in the Original relation. As a result, the two environments

Table 7: **Episode success rate (ESR, %) on both collections.** Weak and strict ESR are reported.

Model	Method	Historical								Temporal							
		Identification				Discovery				Identification				Discovery			
		Original		Counterfactual		Original		Counterfactual		Original		Counterfactual		Original		Counterfactual	
		Weak	Strict	Weak	Strict	Weak	Strict	Weak	Strict	Weak	Strict	Weak	Strict	Weak	Strict	Weak	Strict
DeepSeek-V4-Flash	Prior-only	95.0	88.3	0.0	0.0	51.7	31.7	0.0	0.0	45.8	40.3	5.6	0.0	0.0	0.0	0.0	0.0
	Fixed	96.7	91.7	58.3	48.3	86.7	53.3	16.7	0.0	61.1	51.4	36.1	33.3	45.8	12.5	27.8	9.7
	Adaptive	81.7	76.7	38.3	33.3	48.3	23.3	6.7	0.0	58.3	52.8	51.4	44.4	36.1	9.7	22.2	11.1
K2-Horizon-375B	Prior-only	86.7	83.3	0.0	0.0	41.7	25.0	0.0	0.0	20.8	12.5	9.7	5.6	4.2	0.0	0.0	0.0
	Fixed	83.3	78.3	38.3	36.7	73.3	41.7	10.0	0.0	44.4	40.3	33.3	27.8	31.9	13.9	12.5	5.6
	Adaptive	73.3	73.3	36.7	26.7	56.7	33.3	8.3	0.0	50.0	33.3	38.9	33.3	25.0	12.5	8.3	4.2
Qwen3.6-Plus	Prior-only	41.7	36.7	0.0	0.0	20.0	13.3	0.0	0.0	27.8	23.6	1.4	0.0	2.8	0.0	0.0	0.0
	Fixed	41.7	36.7	1.7	1.7	21.7	18.3	0.0	0.0	30.6	25.0	15.3	13.9	25.0	12.5	13.9	5.6
	Adaptive	1.7	1.7	0.0	0.0	0.0	0.0	0.0	0.0	27.8	25.0	19.4	19.4	9.7	6.9	4.2	2.8
Qwen3.7-Plus	Prior-only	83.3	80.0	0.0	0.0	51.7	31.7	0.0	0.0	43.1	40.3	6.9	0.0	5.6	0.0	0.0	0.0
	Fixed	86.7	81.7	40.0	38.3	78.3	48.3	10.0	0.0	48.6	40.3	33.3	33.3	19.4	13.9	11.1	6.9
	Adaptive	50.0	46.7	15.0	15.0	36.7	21.7	3.3	0.0	38.9	34.7	29.2	27.8	11.1	6.9	9.7	6.9
GPT-5.5	Prior-only	73.3	73.3	31.7	31.7	60.0	35.0	0.0	0.0	9.7	8.3	36.1	19.4	15.3	0.0	0.0	0.0
	Fixed	100.0	95.0	100.0	95.0	100.0	63.3	18.3	0.0	68.1	45.8	63.9	38.9	70.8	19.4	29.2	9.7
	Adaptive	100.0	98.3	95.0	88.3	98.3	63.3	85.0	0.0	81.9	48.6	65.3	41.7	84.7	18.1	68.1	12.5
GPT-5.6 Luna	Prior-only	86.7	83.3	1.7	1.7	51.7	33.3	0.0	0.0	18.1	11.1	8.3	0.0	13.9	0.0	0.0	0.0
	Fixed	95.0	90.0	45.0	40.0	91.7	55.0	15.0	0.0	55.6	40.3	36.1	29.2	54.2	18.1	29.2	11.1
	Adaptive	88.3	83.3	31.7	28.3	75.0	45.0	28.3	0.0	61.1	47.2	30.6	26.4	36.1	11.1	22.2	8.3
GPT-5.6 Terra	Prior-only	93.3	91.7	0.0	0.0	50.0	26.7	0.0	0.0	22.2	9.7	15.3	8.3	11.1	0.0	0.0	0.0
	Fixed	100.0	93.3	95.0	85.0	100.0	60.0	15.0	0.0	59.7	41.7	51.4	38.9	63.9	20.8	30.6	8.3
	Adaptive	95.0	91.7	80.0	73.3	75.0	45.0	35.0	0.0	69.4	55.6	62.5	41.7	26.4	6.9	5.6	1.4
GPT-5.6 Sol	Prior-only	100.0	93.3	1.7	1.7	58.3	36.7	0.0	0.0	23.6	16.7	15.3	5.6	22.2	0.0	0.0	0.0
	Fixed	100.0	98.3	100.0	96.7	100.0	65.0	18.3	0.0	68.1	55.6	63.9	45.8	70.8	20.8	36.1	11.1
	Adaptive	100.0	91.7	95.0	86.7	100.0	63.3	63.3	0.0	91.7	55.6	80.6	55.6	90.3	20.8	73.6	13.9
GPT-6 Astra	Prior-only	98.3	78.3	0.0	0.0	56.7	36.7	0.0	0.0	25.0	8.3	8.3	0.0	23.6	0.0	0.0	0.0
	Fixed	100.0	81.7	100.0	88.3	100.0	66.7	20.0	0.0	77.8	38.9	72.2	37.5	90.3	20.8	41.7	9.7
	Adaptive	100.0	80.0	100.0	85.0	100.0	63.3	95.0	1.7	97.2	48.6	95.8	44.4	100.0	19.4	98.6	16.7

differ in their active terms even though they share the parametric form. Table 8 lists the Historical task families and their mutations.

Temporal sources. The current Temporal collection contains 24 families. Tables 9 gives interventions and sources. Collection batch and publication date are distinct quantities.

D LLM Disclosure

Generative AI tools were used solely to assist in summarizing existing literature and refining the manuscript for grammar and readability. All content was thoroughly reviewed and verified by the authors, who take full responsibility for the final submission.

Table 8: **The 20 Historical task families.** Tasks are deterministic idealizations grounded in cited sources rather than literal replications. Brackets denote valid input domains; $\mathcal{C}$ and $\mathcal{R}$ indicate controllable inputs and observables.

Family	Historical anchor	Experimental interface	Historical relation	Validity scope	Counterfactual lineage	Separating intervention
pendulum	Huygens, <i>Horologium Oscillatorium</i> (1673)	$\mathcal{C}$: length $L \in [0.2, 2]$ m; bob mass $m \in [0.05, 1]$ kg; release angle $\theta \in [1, 10]^\circ$. $\mathcal{R}$: period T (s).	$T = T_{\text{ref}}(L/1 \text{ m})^{0.5} \times (m/0.20 \text{ kg})^0$; $T_{\text{ref}} = 2.006409293$ s, equivalently $g = 9.80665 \text{ m s}^{-2}$; independent of θ .	Small angle; simple pendulum; uniform g ; negligible drag, friction, buoyancy, and finite-amplitude correction.	M1: $m^0 \rightarrow m^{0.25}$; M2: $p = 1.20$; M3: $u_L u_m$ interaction.	Cross length and mass from low to high values while holding angle fixed; include the reference and off-reference angle endpoints.
young_double_slit	Young, Bakerian lecture (read 1803; published 1804)	$\mathcal{C}$: wavelength $\lambda \in [400, 700]$ nm; screen distance $D \in [0.5, 3]$ m; slit separation $d \in [0.1, 1]$ mm. $\mathcal{R}$: fringe spacing Δy (mm).	$\Delta y = 1.1 \text{ mm} \times (\lambda/550 \text{ nm})^1 \times (D/1 \text{ m})^1 \times (d/0.50 \text{ mm})^{-1}$.	Mutually coherent narrow slits; central paraxial fringes; normal aligned screen; fixed slit width and polarization.	M1: $d^{-1} \rightarrow d^{-0.5}$; M2: $p = 0.80$; M3: $u_\lambda u_D$ interaction.	Sweep d across its range, then cross wavelength and screen distance at separated levels.
galileo_inclined_plane	Galileo, <i>Two New Sciences</i> (1638)	$\mathcal{C}$: distance $s \in [0.25, 4]$ m; incline $\theta \in [2, 10]^\circ$; ball mass $m \in [0.05, 0.5]$ kg. $\mathcal{R}$: collected water (g).	$W = 24 \text{ g}(s/1 \text{ m})^{0.5} \times [\sin \theta / \sin 5^\circ]^{-0.5} \times (m/0.20 \text{ kg})^0$.	Similar balls roll from rest without slip at constant acceleration; fixed inertia ratio and steady water-clock flow.	M1: $s^{0.5} \rightarrow s^{0.7}$; M2: $p = 1.20$; M3: $u_s u_\theta$ interaction.	Cross distance and incline over interior and endpoint settings, with mass varied to test the historical invariance.
boyle_j_tube	Boyle, trapped-air compression manuscript (2 October 1661)	$\mathcal{C}$: added pressure $P_{\text{add}} \in [0, 220]$ kPa; thermal condition fixed to room temperature. $\mathcal{R}$: volume (mL), absolute pressure (kPa).	$V = 48 \text{ mL} (P/101.3 \text{ kPa})^{-1}$, with $P = 101.3 \text{ kPa} + P_{\text{add}}$.	Fixed amount of dilute gas; quasistatic isothermal equilibrium; calibrated end-space; no leakage, adsorption, or dissolution.	M1: $V \propto P^{-1} \rightarrow P^{-0.7}$; M2: $p = 0.80$; M3: bounded single-axis curvature $1 + 0.20 u_P (1 - u_P)$.	Sweep added pressure from zero through two interior values to the upper endpoint after thermal equilibration.
photoelectric	Einstein (1905), Millikan's direct experiment (1916), and his Nobel lecture (delivered 1924)	$\mathcal{C}$: frequency $\nu \in [400, 1000]$ THz; absorbed photon flux $\phi \in [0.1, 2]$ relative. $\mathcal{R}$: stopping potential (V), saturation current (μA).	$V_{\text{stop}} = \max(0, a\nu - \Phi/e)$, $a = 4.135667696 \times 10^{-15} \text{ V s}$, $\Phi/e = 2.28 \text{ V}$; above threshold, $I_{\text{sat}} = 3.0\phi \mu\text{A}$.	Clean uniform cathode; monochromatic single-photon regime; complete collection; no contact drift, space charge, multi-photon response, or dark current.	M1: $h/e : 4.135667696 \times 10^{-15} \rightarrow 3.20 \times 10^{-15} \text{ V s}$; M2: $p = 1.20$; M3: $u_\nu u_\phi$ interaction.	Probe below and above the shifted threshold at widely separated frequencies and cross those settings with low and high flux.
faraday_induction	Faraday, <i>Experimental Researches in Electricity</i> , vol. I (experiments 1831 onward; collected 1839)	$\mathcal{C}$: turns $N \in \{50, 100, 200, 400\}$; area $A \in [5, 100] \text{ cm}^2$; $\Delta B \in [1, 50] \text{ mT}$; ramp time $\Delta t \in [5, 500] \text{ ms}$. $\mathcal{R}$: peak emf (mV).	$ \mathcal{E} = 20 \text{ mV}(N/100)^1 \times (A/20 \text{ cm}^2)^1 \times (\Delta B/10 \text{ mT})^1 \times (\Delta t/100 \text{ ms})^{-1}$.	Uniform normal field; constant-slope ramp; identical turns; high-impedance readout; negligible loading, self-inductance, and stray flux.	M1: $\Delta t^{-1} \rightarrow \Delta t^{-0.7}$; M2: $p = 0.80$; M3: $u_N u_A$ interaction.	Sweep ramp duration across decades and cross turns with loop area while holding field change controlled.
radioactive_decay	Rutherford and Soddy (1902, 1903)	$\mathcal{C}$: elapsed time $t \in [0, 300]$ min; active mass $m \in [0.25, 4]$ mg. $\mathcal{R}$: count rate (counts min^{-1}).	$R = 20 + 800(m/1 \text{ mg})^1 \times 2^{-t/(60 \text{ min})}$ counts min^{-1} .	One radionuclide; common preparation and geometry; fixed background and efficiency; negligible self-attenuation and dead time.	M1: $m^1 \rightarrow m^{0.7}$; M2: $p = 1.20$; M3: $u_t u_m$ interaction.	Cross early/late times with low/high masses; retain a zero-time point to isolate mass scaling from decay.
rutherford_scattering	Geiger–Marsden (1909), Rutherford (1911), and Geiger–Marsden (1913)	$\mathcal{C}$: angle $\theta \in [20, 160]^\circ$; $Z \in \{13, 29, 47, 79\}$; target density $n \in [0.1, 4]$ relative; exposure $\Phi \in [0.25, 2]$ relative. $\mathcal{R}$: count rate (counts min^{-1}).	$R = 120 \text{ cpm}(Z/79)^2 \times n^1 \Phi^1 \times [\sin(\theta/2) / \sin 30^\circ]^{-4}$	Elastic nonrelativistic unscreened Coulomb scattering; thin single-scattering target; fixed beam energy and narrow detector acceptance.	M1: angular exponent $-4 \rightarrow -3$; M2: $p = 0.80$; M3: $u_\theta u_Z$ interaction.	Sweep angle broadly and cross it with at least two separated Z values; vary density/exposure to verify retained linear factors.
hooke_spring	Hooke, <i>Lectures de Potentia Restitutiva</i> (1678)	$\mathcal{C}$: force $F \in [0.5, 4]$ N; stiffness $k \in [20, 80] \text{ N m}^{-1}$. $\mathcal{R}$: extension x (cm).	$x = 5 \text{ cm}(F/2 \text{ N}) \times (k/40 \text{ N m}^{-1})^{-1} \times [1 + c((F/2 \text{ N})^2 - 1)]$, with $c = 0$.	Reversible quasistatic tensile regime; negligible hysteresis, creep, inertia, and temperature drift.	M1: cubic correction coefficient $0 \rightarrow 0.20$; M2: $p = 1.20$; M3: $u_F u_k$ interaction.	Sweep force at fixed stiffness to expose cubic curvature, then cross force and stiffness away from their reference values.
torricelli_efflux	Torricelli, <i>Opera geometrica</i> (1644)	$\mathcal{C}$: liquid head $h \in [0.1, 2]$ m; orifice area $A \in [1, 20] \text{ mm}^2$. $\mathcal{R}$: exit speed v (m s^{-1}).	$v = 4.43 \text{ m s}^{-1} \times (h/1 \text{ m})^{0.5} \times (A/5 \text{ mm}^2)^0$.	Steady incompressible lossless efflux; large reservoir; equal surface/outlet pressure; fixed opening shape and negligible contraction loss.	M1: $h^{0.5} \rightarrow h^1$; M2: $p = 0.80$; M3: $u_h u_A$ interaction.	Sweep head from low to high at fixed area, then vary area at matched heads to test both invariance and interaction.

Continued on next page

Table 8 (continued)

Family	Historical anchor	Experimental interface	Historical relation	Validity scope	Counterfactual	Separating intervention
ohm_circuit	Ohm, <i>Die galvanische Kette, mathematisch bearbeitet</i> (1827)	C : voltage $V \in [0.2, 5]$ V; resistance $R \in [20, 500] \Omega$. $\mathcal{R}$: current I (mA).	$I = 10 \text{ mA} (V/1 \text{ V})^1 \times (R/100 \Omega)^{-1}$.	Steady DC at fixed temperature; ideal source and meter; negligible self-heating, contact resistance, and parasitics.	M1: $R^{-1} \rightarrow R^{-0.5}$; M2: $p = 1.20$; M3: $u_V u_R$ interaction.	Sweep resistance over its full range at fixed voltage and cross low/high voltage with low/high resistance.
rc_discharge	Thomson's condenser-discharge analysis (1853); the benchmark uses its ideal zero-inductance RC limit	C : $V_0 \in [2, 10]$ V; $R \in [25, 100] \text{ k}\Omega$; $C \in [10, 40] \mu\text{F}$; $t \in [0.05, 4]$ s. $\mathcal{R}$: capacitor voltage (V).	$V(t) = V_0 \exp[-a(t/RC)^q]$, with $a = 1, q = 1$, and $RC = R_{\text{k}\Omega} C_{\mu\text{F}}/1000$ s.	Ideal lumped first-order RC; constant components; high-impedance meter; negligible leakage, dielectric absorption, and parasitics.	M1: normalized-time shape $t^1 \rightarrow t^2$; M2: $p = 0.80$; M3: $u_{V_0} u_R$ interaction.	Sample several normalized times $t/(RC)$, then cross initial voltage and resistance while separately varying capacitance.
stefan_boltzmann	Stefan (1879) and Boltzmann (1884)	C : emitter $T_e \in [400, 900]$ K; ambient $T_a \in [250, 350]$ K; area $A \in [20, 200] \text{ cm}^2$. $\mathcal{R}$: net radiative power (W).	$P = 35.439840119 \text{ W} \times (A/100 \text{ cm}^2) \times [(T_e/500 \text{ K})^4 - (T_a/500 \text{ K})^4]$.	Diffuse black emitter in a black enclosure with view factor one; uniform temperatures; calibrated exclusion of non-radiative transfer.	M1: linked $T_e^4, T_a^4 \rightarrow T_e^3, T_a^3$; M2: $p = 1.20$; M3: $u_{T_e} u_{T_a}$ interaction.	Sweep emitter temperature at fixed ambient, vary ambient at matched emitter temperature, and check area proportionality.
newton_cooling	Newton, <i>Scala graduum caloris</i> (1701)	C : $T_0 \in [50, 100]^\circ\text{C}$; $T_a \in [10, 30]^\circ\text{C}$; $t \in [0, 25]$ min. $\mathcal{R}$: object temperature ($^\circ\text{C}$).	$T - T_a = (T_0 - T_a)e^{-kt}$, with $k = 0.12 \text{ min}^{-1}$ and fast-mode weight $w = 0$.	Lumped capacitance; constant ambient, transfer coefficient, and properties; negligible sensor lag and unmodelled loss channels.	M1: fast-mode weight $0 \rightarrow 0.25$; M2: $p = 0.80$ on ΔT ; M3: $u_{T_0} u_{T_a}$ modulation of ΔT only.	Resolve the early-time curve at fixed endpoints, then cross initial and ambient temperatures while comparing temperature excess.
beer_lambert	Bouguer (1729), Lambert (1760), and Beer (1852)	C : incident power $I_0 \in [20, 100] \mu\text{W}$; concentration $c \in [0.05, 1] \text{ mM}$; path $l \in [0.25, 2] \text{ cm}$. $\mathcal{R}$: transmitted power (μW).	$I = I_0 e^{-\alpha c(l/1 \text{ cm})^q}$, with $\alpha = 0.6 \text{ mM}^{-1}$ and $q = 1$.	Homogeneous stable non-scattering, non-fluorescent absorber in the linear regime; fixed wavelength, solvent, temperature, and calibrated windows.	M1: path exponent $l^1 \rightarrow l^2$; M2: $p = 1.20$; M3: $u_{I_0} u_c$ interaction.	Sweep path length at fixed concentration, then cross incident power and concentration at a shared path length.
michaelis_menten	Michaelis and Menten (1913); Haldane (1930)	C : substrate $S \in [0.05, 10] \text{ mM}$; enzyme $E \in [0.25, 2]$ relative. $\mathcal{R}$: initial rate ($\mu\text{mol L}^{-1} \text{ min}^{-1}$).	$v = V_{\text{max}} ES/[K_m + S + \gamma S^2/K_m]$, with $V_{\text{max}} = 80 \mu\text{mol L}^{-1} \text{ min}^{-1}$, $K_m = 1 \text{ mM}$, and $\gamma = 0$.	Single-substrate quasi-steady initial-rate window; negligible depletion, product, reverse reaction, inactivation, inhibition, and mass-transfer limitation.	M1: substrate-inhibition strength $0 \rightarrow 0.25$; M2: $p = 0.80$; M3: $u_S u_E$ interaction.	Sweep substrate through the rising, saturation, and high-substrate regimes; cross enzyme levels at shared substrate values.
coulomb_force	Coulomb, first memoir (presented 1785; cited from the 1789 collected volume)	C : charges $q_1, q_2 \in [1, 10] \text{ nC}$; separation $r \in [2, 20] \text{ cm}$. $\mathcal{R}$: force magnitude (μN).	$F = 22.47 \mu\text{N} \times (q_1/5 \text{ nC})^1 \times (q_2/5 \text{ nC})^1 \times (r/10 \text{ cm})^{-2}$.	Static point charges in vacuum; fixed localized distributions; no screening, polarization, image forces, or external fields.	M1: $r^{-2} \rightarrow r^{-1}$; M2: $p = 1.20$; M3: $u_{q_1} u_{q_2}$ interaction.	Sweep separation over its range, then independently cross both charges at a fixed separation.
kepler_orbit	Kepler, <i>Harmonices mundi libri V</i> (1619)	C : semi-major axis $a \in [0.2, 8] \text{ AU}$; eccentricity $e \in [0, 0.7]$. $\mathcal{R}$: orbital period (days).	$P = 365.25 \text{ days} \times (a/1 \text{ AU})^{1.5} (1+e)^0$.	Bound Newtonian two-body orbit about the same fixed central mass; negligible test mass, perturbations, tides, drag, and relativistic corrections.	M1: $a^{1.5} \rightarrow a^1$; M2: $p = 0.80$; M3: $u_a u_e$ interaction.	Sweep semi-major axis across more than one decade and vary eccentricity at a shared off-reference axis.
brownian_diffusion	Einstein, Brownian-motion paper (1905)	C : lag $t \in [0.05, 5]$ s; tracer radius $r \in [0.2, 1] \mu\text{m}$. $\mathcal{R}$: two-dimensional MSD (μm^2).	$\text{MSD}_{2D} = 8 \mu\text{m}^2 \times (t/1 \text{ s})^1 \times (r/0.5 \mu\text{m})^{-1}$.	Overdamped dilute spheres in a Newtonian fluid; fixed temperature and viscosity; far from walls; lag below drift/confinement times.	M1: $t^1 \rightarrow t^2$; M2: $p = 1.20$; M3: $u_t u_r$ interaction.	Sweep lag over two decades at fixed radius, then cross lag and radius at separated levels.
poiseuille_flow	Hagen (1839) and Poiseuille (preliminary reports 1840–1841; memoir issued 1844 and collected 1846)	C : $\Delta P \in [2, 10] \text{ kPa}$; radius $r \in [0.25, 0.5] \text{ mm}$; length $L \in [20, 50] \text{ cm}$; viscosity $\mu \in [1, 3] \text{ mPa s}$. $\mathcal{R}$: flow rate (mL s^{-1}).	$Q = 1.2271846303 \text{ mL s}^{-1} \times (\Delta P/10 \text{ kPa})^1 \times (r/0.5 \text{ mm})^4 \times (L/20 \text{ cm})^{-1} \times (\mu/1 \text{ mPa s})^{-1}$.	Steady fully developed incompressible Newtonian laminar flow; rigid straight circular tube; no-slip wall; losses measured only between pressure taps.	M1: $r^4 \rightarrow r^3$; M2: $p = 0.80$; M3: $u_{\Delta P} u_r$ interaction.	Sweep radius at fixed pressure, length, and viscosity; cross pressure and radius, then verify retained inverse length/viscosity factors.

Table 9: **The 24 families in the Temporal collection.** These are derived from 23 primary papers and paired with benchmark-authored counterfactual responses. The source-reported branch is a deterministic implementation of the selected relation under the stated approximations. It does not reproduce all experiments or raw data in the paper. Numerical intervals and discrete sets below are the current legal query domains.

Family	Journal and source	Experimental interface	Source-reported relation	Validity scope	Counterfactual construction	Separating queries
Defect super-symmetric entropy	<i>Physical Review Letters</i> (Huang et al., 2026b).	C inverse Rényi index $\gamma \in [0.5, 2]$. $\mathcal{R}$ normalized entropy y (dimensionless).	$y = 12[1 + 0.125(\gamma - 1)]$.	Symmetric $k = 2$ Wilson surface in the $A_3(2, 0)$ theory. $r_1 = r_2 = 1/2$. Exact relation at fixed theory and representation.	M1 replaces γ by γ^2 , preserving $y(1) = 12$.	Query $\gamma = 0.5$ and 2, and retain $\gamma = 1$ as a common reference.
Turbulent scaling exponent	<i>Physical Review Letters</i> (Mailybaev and Thalabard, 2026).	C moment order $p \in [1.4, 4.6]$. $\mathcal{R}$ scaling exponent ζ_p (dimensionless).	$\zeta_p = p/3 - Cp(p-2)\epsilon^2$, $C = (2^{2/3} + 1)/(4 \cdot 2^{1/3} \ln 2)$ and $\epsilon = 0.1$.	Inertial-range shell-model asymptotics through $O(\epsilon^2)$. Terms $O(\epsilon^4)$ are omitted. Continuous moment orders are benchmark queries.	M1 changes ϵ from 0.1 to 0.2. The correction is multiplied by four and $\zeta_2 = 2/3$ is retained.	Sample moment orders below and above $p = 2$, including 1.4 and 4.6. The protected $p = 2$ alone does not separate the laws.
MnF ₂ crossover displacement	<i>Physical Review Letters</i> (Ohlendorf et al., 2026).	C strain $\epsilon_{xy} \in \{-0.0004, -0.0008, -0.0011, -0.0015\}$ and field $B \in [0.644, 5.796]$ T. $\mathcal{R}$ crossover shift ΔT^* (K).	$T_c = 67.467 - 59\epsilon_{xy} - 0.01585B^2$. $\Delta T^* = T_c[0.56(0.67171381472) \cdot (-B\epsilon_{xy})/T_c]^2/3$.	Compression along $[110]$, field along c . Mean-field exponent 2/3. All admitted strains are negative, so $-B\epsilon_{xy} = B\epsilon_{xy} $.	M1 replaces every B by $W_{1-\epsilon_{xy}/0.0015}(B)$ including its occurrences in T_c .	Cross strain levels with interior fields, including $B = 3.22$ T. Endpoint fields are preserved by the composition.
Confined ciliate swimming	<i>Physical Review Letters</i> (Antunes et al., 2026).	C mean slip $u \in [-0.4, 0.4]$ and amplitude $a \in [0.24, 1.36]$. $\mathcal{R}$ normalized laboratory-frame mean swimming velocity $\langle v_p \rangle / v_m$. All dimensionless.	$\langle v_p \rangle / v_m = \frac{1}{\sqrt{\max[(1-u)^2 - a^2, 0]}}$.	Resonant channel and metachronal wavelengths, lubrication and incompressible Stokes flow. Fixed reduced mobility $ G_n = 2$. Includes the phase-locked plateau.	M1 and M2 replace u and a by $W_2(u)$ and $W_2(a)$.	Cross interior slip and amplitude settings through the oscillatory-to-locked transition. Include interior values because corners alone are preserved.
Quantum interconnect rate	<i>Physical Review Letters</i> (Luo et al., 2026).	C modes $M \in \{2, 4, 8, 11\}$, storage efficiency $\eta_s \in [0.54, 0.86]$ and noise $p_n \in \{0.01, 0.10, 0.50\}$. $\mathcal{R}$ scaled rate $\tilde{R} = (L_0/c)R_{\text{qm}}$ (bits).	$A = 1 - (M-1)p_n/M$. $\mathcal{I} = \log_2 M + A \log_2 A + (M-1)(p_n/M) \log_2(p_n/M)$. $\tilde{R} = 0.7e^{-250/44} \mathcal{I} \eta_s^4 / [27(2 - \eta_s)(4 - 3\eta_s)]$.	Ideal 1000-km, four-segment repeater with two nesting levels. $N = \eta_d = 1$, $L_{\text{att}} = 22$ km. Equal input probabilities and depolarizing memory noise. Lifetime losses omitted.	M1 replaces all η_s occurrences by $W_{1+p_n/0.1}(\eta_s)$, retaining the same information term.	Cross noise and mode levels at interior $\eta_s = 0.7$. Efficiency endpoints alone are preserved.
Time-dependent CP asymmetry	<i>Physical Review Letters</i> (LHCb Collaboration, 2026).	C proper time $t \in [1.25, 7.25]$ ps. $\mathcal{R}$ integrated CP asymmetry A_{CP} (dimensionless).	$A_{CP}(t) = 0.82 \sin(0.507t) + 0.13 \cos(0.507t)$.	Helicity-angle-integrated $B^0 \rightarrow K_S^0 \mu^+ \mu^-$ asymmetry. $\Delta \Gamma_d = 0$ approximation and fixed dilepton-mass exclusions. Proper time is an observation coordinate.	M1 replaces both time occurrences by $W_2(t)$.	Sample interior proper times across the oscillation, in addition to 1.25 and 7.25 ps. The two endpoints alone coincide.
Fermentation minimum selling price	<i>Nature Communications</i> (Bhagwat et al., 2026).	C yield $Y \in [0.28, 0.92]$ g product/g sugar and titer $T \in [21, 109]$ g product/L. $\mathcal{R}$ minimum selling price P (2019 USD/kg product).	$P = [0.311 + 0.353/Y + 10.9/T + 0.356/(YT)]/0.983$.	Fitted techno-economic relation for glucose-to-succinic acid, low-pH index 25. Productivity $0.66 \text{ g L}^{-1} \text{ h}^{-1}$ and recovery 0.983. Large-scale fed-batch plant model.	M1 and M2 replace yield Y and titer T by $W_2(Y)$ and $W_2(T)$ in every occurrence.	Cross interior yield and titer levels with their endpoints to distinguish both reciprocal terms and their interaction.
Shear-thinning cutting force	<i>Nature Communications</i> (Zhao et al., 2026).	C $r \in \{0.6856, 0.9775, 1.0829, 1.4455, 1.5566\}$ μm . $v \in \{0.083, 0.125, 0.167, 0.208, 0.250, 0.292, 0.333, 0.375, 0.417, 0.458, 0.500\}$ m/s. $h \in \{5, 23, 33\}$ mm. $\mathcal{R}$ relative force $y = F/F_{\text{ref}}$ (dimensionless).	$y = (r/0.6856)^{0.968} \cdot (v/0.5)^{0.72} \cdot (h/33)^{1.244}$.	Printed empirical fit for the fixed optimized starch fluid and calibrated straight blades. Discrete joint query grid. Other compositions, blade geometries and the unlabelled 16-mm-sheet series are excluded.	M1 replaces the radius exponent 0.968 by $0.968(v/0.5)$, coupling radius and speed. The common normalization is retained.	Cross at least two nonreference radii with low and high speeds, with depth fixed. The reference radius or reference speed alone hides the mutation.

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Table 9 (continued)

Family	Journal and source	Experimental interface	Source-reported relation	Validity scope	Counterfactual construction	Separating queries
Off-diagonal scattering density	<i>Physical Review Letters</i> (Köhnes et al., 2026).	$\mathcal{C}$ cross section $\sigma \in [0.04, 0.36]$. $\mathcal{R}$ local probability-density value p_σ . Both coordinates are dimensionless.	$m = 0.967^2/(2(0.967) + 7.013)$, $z = \sigma/m$, $a = (2(0.967) - 1)/(2(0.967) + 7.013)$. $p_\sigma = e^{-z}[1 + a((4 - z)z - 2)]/m$.	Leading and first subleading Heidelberg asymptotic density for $\beta = 2$, $T_1 = T_2 = 0.967$ and absorption 7.013. Used near the Ericson transition. The finite query interval is not a separately normalized support.	M1 composes p_σ with $W_2(\sigma)$. M2 multiplies it by $W_2'(\sigma) = 2(\sigma - 0.04)/0.32$.	Query endpoints and interior cross sections. The Jacobian changes density heights even where the coordinate composition fixes an endpoint.
Thin-film differential reflectance	<i>Physical Review Letters</i> (Puri et al., 2026).	$\mathcal{C}$ SnS thickness $d \in [72.05, 252.45]$ nm. $\mathcal{R}$ AC-polarized differential reflectance $\Delta R/R$ (dimensionless).	$R = (r_{01} + r_{12}z)/(1 + r_{01}r_{12}z) ^2$, $z = e^{4\pi i \tilde{n}d/800}$, $r_{01} = (\tilde{n} - \tilde{n})/(1 + \tilde{n})$, $r_{12} = (\tilde{n} - 1.7276)/(\tilde{n} + 1.7276)$. $\Delta R/R = -0.00302 \partial_n \ln R$ at $\tilde{n} = n + 0.12i$, $n = 4.23$.	Normal-incidence 800-nm light in air/SnS/MgO. First-order real-index perturbation only, with imaginary-index change omitted. All multiple reflections and the analytic derivative are retained.	M1 replaces every thickness occurrence by $W_2(d)$ in the complete differentiated Fresnel response.	Sweep interior thickness through interference extrema. The original thickness endpoints are common to both laws.
Squeezed-vacuum energy surface	<i>Physical Review Letters</i> (Ramírez et al., 2026).	$\mathcal{C}$ bosonic coordinates $q, p \in [-1.6, 1.6]$. $\mathcal{R}$ thermodynamic-limit energy h_∞ in the fixed model energy unit.	$h_\infty = (q^2 + p^2)/2 - 2\sqrt{1 + 0.04q^2(q^2 + p^2)}$.	Mean-field squeezed-vacuum surface after eliminating atomic coordinates. $\omega = 1$, $\omega_0 = 2$, $\gamma/\omega = 0.4$, below spectral collapse. Not an exact finite-system energy.	M1 and M2 replace all q and p occurrences by $W_2(q)$ and $W_2(p)$.	Cross interior positive, zero and negative phase-space coordinates to test curvature and the source reflection symmetries.
Pulsed magnetic-field sensing	<i>Physical Review X</i> (Tabatabaei et al., 2026).	$\mathcal{C}$ field $B \in [10^{-8}, 9 \times 10^{-8}]$ T and pulse blocks $N \in \{10, 30, 58, 80, 100\}$. $\mathcal{R}$ signal-to-noise ratio.	$t_c = 1.78 \times 10^{-6} N$ s. $A_N = \sqrt{60/(t_c + 0.004)} \times e^{-t_c/0.000104}/\mathcal{N}$. $\text{SNR} = A_N \sin(0.34\gamma_e B t_c)$, $\gamma_e = 176086268505.04355$ s ⁻¹ T ⁻¹ and $\mathcal{N} \simeq 14.109773394162719$.	Fixed acquisition and noise parameters of the nonlinear sensitivity model. Sinusoidal field at a filter-function lobe and sensing duration much shorter than T_1 . $\mathcal{N}$ is evaluated from the frozen source noise expression.	M1 replaces B only inside the phase by $W_{1+N/100}(B)$. Acquisition, decay and noise terms retain their forms.	Cross interior field strengths with distinct block counts. Field endpoints alone conceal the composition.
Undetected-light visibility	<i>Nature Communications</i> (Severs Millard et al., 2026). Advance publication.	$\mathcal{C}$ object-arm intensity transmission $T \in [0.1, 0.9]$. $\mathcal{R}$ fringe visibility V . Both dimensionless.	$a = 0.784^2(0.47)T$. $b = 0.8(0.118)$ and $c = 0.885^2 + 0.8$. $V = \frac{2\sqrt{0.8a(0.885 + \sqrt{b})}}{b + c(1 + a)}$.	Michelson nonlinear thin-film model with undepleted pump and seed. Stimulated emission dominates. Only the first four-wave-mixing reflection and within-window interference are retained.	M1 replaces both transmission occurrences by $W_2(T)$.	Scan interior transmission values between 0.1 and 0.9. The two endpoints alone agree.
Davemaote phase boundary	<i>Nature Communications</i> (Hao et al.).	$\mathcal{C}$ pressure $P \in [36, 124]$ GPa. $\mathcal{R}$ transition temperature T_{tr} (K).	$T_{tr} = 533 + 2.4P$.	Printed fitted tetragonal-to-cubic phase boundary for lower-mantle davemaote. Deterministic use of central fit coefficients. Fit uncertainties are not per-query noise.	M1 replaces P by $W_2(P)$.	Compare at least two interior pressures against the source straight line, with endpoints as shared references.
CoO orbital magnetoresistance	<i>Science</i> (Schmitt et al., 2026).	$\mathcal{C}$ remaining Cu* thickness $t \in [3.88, 5.32]$ nm. $\mathcal{R}$ orbital-magnetoresistance amplitude A (percent).	$A = 0.32 + 0t$ percent.	Approximate plateau of CoO(5 nm)/Cu* at 150 K. One Hall bar is progressively ion-milled and exposed to air for 12 h after each etch. Thickness is resistance-calibrated.	M1 changes the affine slope from 0 to 0.08 percent/nm, giving $A = 0.32 + 0.08t$.	Compare low and high remaining thickness to test the plateau against a nonzero slope under the fixed protocol.

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Table 9 (continued)

Family	Journal and source	Experimental interface	Source-reported relation	Validity scope	Counterfactual construction	Separating queries
MnF ₂ transition baseline	<i>Physical Review Letters</i> (Ohlendorf et al., 2026).	$\mathcal{C}$ loading-path strain $\epsilon \in \{-0.0015, -0.0011, -0.0008, -0.0004\}$ and field $B \in \{0, 6.44\}$ T. $\mathcal{R}$ baseline T_c (K).	$T_c = 67.467 - 59\epsilon - 0.01585B^2$.	Fitted non-alternating magnetic baseline along compressive [110] loading, including co-induced symmetric strain. Not independently imposed pure shear or the shifted crossover T^* .	M1 replaces B^2 by B^4/B_0^2 , with fixed $B_0 = 1$ T. The field coefficient retains units K/T ² .	Compare $B = 0$ and 6.44 T at matched strain, then repeat across strain levels.
Quasiperiodic localization exponent	<i>Physical Review Letters</i> (Lin et al., 2026a).	$\mathcal{C}$ comprises couplings $\lambda_1, \lambda_2 \in [0.20, 0.67]$. $\mathcal{R}$ is the Lyapunov exponent L . All quantities are dimensionless.	$\Lambda = \frac{\lambda_2(1 + \sqrt{1 - \lambda_1^2})}{\lambda_1(1 + \sqrt{1 - \lambda_2^2})}$. $L = \max(0, \ln \Lambda)$.	The source-theory unitary almost-Mathieu relation is evaluated at non-Hermiticity $\eta = 0$, irrational quasiperiodic field $\Phi = (\sqrt{5} - 1)/2$ and phase $\theta = 0$. Positive L gives inverse localization length. A zero value covers extended or critical states. The interface returns the asymptotic theoretical exponent without finite-time photon-counting fluctuations.	M1 and M2 replace every occurrence of λ_1 and λ_2 by $W_2(\lambda_1)$ and $W_2(\lambda_2)$.	Cross distinct interior couplings with $\lambda_2 > \lambda_1$, including (0.25, 0.50). Include the reversed pair as a zero-response control. Equal couplings and domain corners alone do not separate the branches.
Viscous-inertial suspension rheology	<i>Physical Review Letters</i> (Konidena et al., 2026).	$\mathcal{C}$ is the Stokes number $St \in [0.01, 300]$. $\mathcal{R}$ is normalized macroscopic friction $r = \mu/\mu_c$. Both are dimensionless.	$r = \frac{1 + 14.3\sqrt{0.065(1 + 0.029 St)}}{1 + 0.1 St}$.	The source-reported dense-suspension relation is restricted to interparticle friction $\mu_f = 0.15$ and the constructed packing-related slice $K_\phi = J + 0.1I^2 = 0.065$, with $St = I^2/J$. The displayed coefficients specify the fixed deterministic response. Changing Stokes number does not independently scan packing or particle friction, and the admitted interval is not extrapolated.	M1 replaces both Stokes-number occurrences by $W_2(St)$.	Sweep logarithmically separated interior values such as 0.1, 10 and 100. Compare the crossover curvature and use the shared endpoints only as controls.
Splashing-drop rim response	<i>Physical Review Letters</i> (Riboux and Gordillo, 2026).	$\mathcal{C}$ is $We \in [288, 592]$. $\mathcal{R}$ comprises rim speed $v_t = V_t/V$ and rim diameter $h_t = H_t/R_d$. All quantities are dimensionless.	$Re = \sqrt{We}/0.0105$, $A = \sqrt{3}/(2Re)$, $b = 1/We$. Let $x > 0$ solve $x^4 - bx - A = 0$ and set $t_e = x^2$. $v_t = 0.45\sqrt{3}/t_e$, $h_t = 2.2t_e^{3/2}$.	The calibrated low-viscosity, high-Reynolds-number approximation describes the early two-jets spray regime. The fixed layer depth is $H/R_d = 1.8$, Ohnesorge number 0.0105, and gas density and viscosity ratios 0.001 and 0.018. Both readouts describe the toroidal rim at $t_0 = 0.07$ in units of R_d/V , including ballistic rim motion and early thickening.	M1 replaces every Weber-number occurrence by $W_2(We)$ in the linked root construction and both outputs.	Sample several interior Weber numbers and compare the coupled speed and diameter curves. Preserve the low and high endpoints as common references rather than using them alone.

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Table 9 (continued)

Family	Journal and source	Experimental interface	Source-reported relation	Validity scope	Counterfactual construction	Separating queries
Trapped-ion optical Magnus response	<i>Physical Review Letters</i> (Leindecker et al., 2026).	$\mathcal{C}$ comprises focal-plane positions $x, y \in [-1.04, 1.04] \mu\text{m}$ and transition category $q \in \{-1, +1\}$. $\mathcal{R}$ is ground-state survival probability P_q , dimensionless.	$g_q = (x/w)(1 + qy/z_R)e^{-(x^2+y^2)/w^2}, w = 1.3 \mu\text{m}, z_R = \pi w^2/(0.729 \mu\text{m}).$ $a = \sqrt{1 + 2(w/z_R)^2}, g_* = \frac{a+1}{2\sqrt{2}e}e^{-(w/z_R)^2/(a+1)^2}.$ $P_q = \cos^2[\pi g_q /(2g_*)].$	A resonant single pulse addresses a fixed $^{40}\text{Ca}^+$ ion initially in $m_g = +1/2$. The beam propagates along z , is polarized along y , and the quantization field points along $-x$. The first correction beyond the paraxial quadrupole-gradient model is retained. Each transition uses its modeled peak coupling over the one-waist square for its own fixed π -pulse calibration.	M1 replaces both occurrences of x by $W_2(x)$. The coordinate y , discrete transition category and fixed pulse calibration are retained.	Cross interior horizontal positions, especially $x = 0$, with positive and negative vertical positions and both transition categories. Track the survival-profile displacement while retaining the horizontal endpoints as controls.
Parity-prepared oscillator squeezing	<i>npj Quantum Information</i> (Lin et al., 2026b). Advance publication.	$\mathcal{C}$ comprises coherent-displacement amplitude $a \in [0.14, 1.26]$ and per-step intensity transmissivity $\eta \in [0.865, 0.985]$, both dimensionless. $\mathcal{R}$ is final conditional squeezing S in dB.	$\Pi_\beta = D(\beta)[1 + (-1)^{\hat{n}}]D^\dagger(\beta)/2, \mathcal{M}_\beta(\rho) = \mathcal{L}_\eta(\Pi_\beta \rho \Pi_\beta)$ with full pure loss $\mathcal{L}_\eta$. Normalize $\rho_* \propto \mathcal{M}_0 \mathcal{M}_{-ia} \mathcal{M}_{ia}(0\rangle\langle 0)$. $S = -10 \log_{10}[4 \min_\theta \text{Var}_{\rho_*}(x_\theta)]$ $x_\theta = (\hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta})/2.$ The card gives the exact nine-exponential reduction.	Three ideal displaced-parity measurements start from vacuum and retain only the all-even outcomes. Full unobserved pure loss follows each measurement, including the third, and the final state is normalized by its full success probability. The response retains coherences and all loss events. The displacement amplitude is not time, and no Gaussian-state or single-loss approximation is used.	M1 replaces all 19 amplitude occurrences by $a^2/1.4$. This uses the preregistered source amplitude reference, not W_2 on the smaller public interval.	Cross low, middle and high admitted amplitudes with several transmissivities. Compare the complete squeezing curve. The public amplitude endpoints are not fixed by this composition.
Mean-field microcanonical coexistence	<i>Physical Review X</i> (Serbyn et al., 2026).	$\mathcal{C}$ comprises $h_x \in [0.06, 0.14]$ and $h_z \in [-0.24, -0.16]$. $\mathcal{R}$ comprises crossing energy density $\epsilon_\times$, common magnetization magnitude $m_\times$ and signed branches m_z^-, m_z^+ . All are dimensionless in the stated Pauli and J_3 units.	$s = -\sqrt{(-h_z + \sqrt{3}h_x)/3}.$ $\epsilon_\times = 2s^3, m_\times = \frac{2}{3}\sqrt{-3h_z + \sqrt{3}h_x}.$ $m_z^\pm = \frac{s \pm \sqrt{s^2 - 8h_z/3}}{2}.$	The uniform thermodynamic LMG-3 mean-field problem maximizes microcanonical entropy at fixed energy $\epsilon = z^3 + h_z z + h_x x$ over $x^2 + z^2 \leq 1$. The selected negative-energy crossing has two global entropy maxima with opposite signed longitudinal magnetizations and the same magnitude. The Pauli magnetization convention is used. This target excludes finite-size disorder, tunneling rates and dynamical ETH hybridization thresholds.	M1 replaces every h_x occurrence in all four outputs by $0.05 + (h_x - 0.05)^2/0.10$. The fixed references are the original safety-box endpoints 0.05 and 0.15.	Cross transverse and longitudinal fields over the legal box. Compare the crossing energy and both signed branches together. The current public transverse-field endpoints do not remain invariant.

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Table 9 (continued)

Family	Journal and source	Experimental interface	Source-reported relation	Validity scope	Counterfactual construction	Separating queries
Loss-mediated transfer and entanglement	<i>Physical Review A</i> (Ren and Li, 2026).	$\mathcal{C}$ is scaled time $\tau = \kappa t \in [30, 270]$. $\mathcal{R}$ comprises $P_{L L}, P_{R L}, C_L, P_{L R}, P_{R R}, C_R$, all dimensionless. The suffix specifies the independent initial qubit excitation, and C_d is reduced two-qubit concurrence.	$u = 2^{-1/4}/10, \mathbf{a}^{(d)} = e^{-iH_{\text{eff}}\tau} \mathbf{e}_d$. $H_{\text{eff}} = \begin{pmatrix} 0 & 0 & u & u \\ 0 & 0 & u & iu \\ u & u & -\frac{1+i}{2} & 0 \\ u & -iu & 0 & \frac{1-i}{2} \end{pmatrix}$ in basis (L, R, c_1, c_2) . $P_{j d} = a_j^{(d)} ^2$, $C_d = 2 a_L^{(d)} a_R^{(d)} $. Amplitudes are unnormalized and lost probability occupies the vacuum.	The engineered sideband-selected rotating-wave model retains both equal-loss cavities with detunings $\Delta_{1,2}/\kappa = \pm 1/2$ and relative coupling phase $-\pi/2$. Each reset starts with one specified qubit excited and both cavities in vacuum. Cavity recycling is unconditional. Intrinsic qubit decoherence and thermal excitation are omitted. No cavity elimination or exactly vanishing reverse transfer is imposed.	M1 replaces all 48 time occurrences in the six complete modal outputs by $W_2(\tau)$. The populations and concurrence require no derivative factor.	Sample several interior times from both preparations, including early and late transients. Compare transferred populations and concurrence jointly. The interval endpoints remain shared.
Non-Markovian cooperative radiation	<i>Quantum</i> (González and Rivas, 2026).	$\mathcal{C}$ is scaled time $\tau = \gamma_0 t \in [2, 18]$. $\mathcal{R}$ comprises unconditional Dicke populations P_2, P_1, P_0 and signed atomic energy-loss rate $j = I/(\omega_0 \gamma_0)$. All four are dimensionless.	$\partial_\tau \rho = -i[H, \rho] + \mathcal{D}[b]\rho$, $H = (J_- b^\dagger + J_+ b)/\sqrt{2}$, $\rho(0) = ee, 0\rangle\langle ee, 0 $. $\mathcal{D}[b]\rho = b\rho b^\dagger - \{b^\dagger b, \rho\}/2$. $P_n = \langle n \text{Tr}_b \rho n \rangle$ for symmetric Dicke states $n = 0, 1, 2$. $j = -\partial_\tau (2P_2 + P_1)$. The card retains every modal coefficient.	Two identical atoms have common coupling phase, a vacuum Lorentzian reservoir, linewidth $\lambda/\gamma_0 = 1/2$ and transition ratio $\omega_0/\gamma_0 = 1000$. The resonant rotating-wave and narrowband pseudomode model retains all reachable excitation sectors and loss recycling. The initial atomic state is fully excited. Negative signed intensity represents reabsorption, and the three populations sum to one.	M1 composes all populations and signed intensity with $g(\tau) = W_2(\tau)$. M2 multiplies the composed intensity by $g'(\tau) = (\tau - 2)/8$, preserving its derivative definition.	Sample interior early and late times and compare populations with signed energy flow. Endpoint populations agree between branches, whereas endpoint intensities need not because of the derivative factor.